The Perron–Frobenius Theorem: From Positive Matrices to Markov Chains and Google's PageRank Algorithm

Mathematical Foundations, Numerical Methods, and Applications

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Abstract

The Perron–Frobenius theorem establishes fundamental properties of the eigenvalues and eigenvectors of matrices with positive or nonnegative entries. In particular, it provides conditions under which a matrix possesses a dominant positive eigenvalue and an associated eigenvector whose components are strictly positive.

These results have important applications in probability theory, numerical linear algebra, Markov chains, and ranking algorithms.

This note develops the Perron–Frobenius theorem for strictly positive matrices, explores its extensions to nonnegative matrices, and investigates its connection to stochastic matrices and stationary probability distributions.

The mathematical framework is then applied to Google's PageRank algorithm, including the modifications introduced to ensure well-defined rankings and desirable convergence properties.

Finally, numerical implementations based on power iteration are presented to illustrate the theoretical results and their practical significance.

Contents

1 Mathematical Preliminaries **4**

1.1 Positive and Nonnegative Matrices 4

1.2 The Action of a Positive Matrix on Nonnegative Vectors 5

1.3 Eigenvalues and the Spectral Radius 6

1.4 Compactness and the Extreme Value Theorem 6

1.5 The Standard Simplex 7

2 The Perron–Frobenius Theorem **12**

2.1 Statement of the Theorem 12

2.2 Existence of a Positive Eigenvalue and Eigenvector 12

2.2.1 Normalization of the Vector 13

2.2.2 Existence of the Maximum 14

2.2.3 Why the Inequality Must Become an Equality 18

2.2.4 Strict Positivity of the Eigenvector 20
2.3 The Spectral Radius and Dominance 20
2.4 Strict Dominance of the Perron Eigenvalue 22
2.5 Algebraic Simplicity 27
2.5.1 Geometric Multiplicity 27
2.5.2 Ruling Out Generalized Eigenvectors 28
2.6 Summary of the Proof 30

3 Stochastic Matrices and Stationary Probability Distributions 31
3.1 What Is a Stochastic Matrix? 31
3.2 Probability Distributions as Vectors 33
3.3 How a Stochastic Matrix Updates a Probability Distribution 33
3.4 Why Is the Eigenvalue One Special? 35
3.5 Stationary Probability Distributions 39
3.6 A Simple Example 39
3.7 Existence of Stationary Distributions 41
3.8 Why Positivity Matters: Uniqueness and Convergence 42
3.9 The Perron–Frobenius Connection 43
3.10 Connection to Web-Page Ranking 44

4 From Hyperlinks to Google's PageRank Algorithm 45
4.1 Representing the Web as a Directed Graph 45
4.2 An Example with Three Web Pages 46
4.3 The Simplest Ranking Method: Counting Incoming Links 46
4.4 The Weighted Linking Matrix 47
4.5 Why Weighted Votes Are Still Not Enough 49
4.6 Deriving the PageRank Eigenvalue Equation 49
4.7 Computing the Rankings: A Three-Page Example 50
4.8 The Random-Surfer Interpretation 52
4.9 Limitations of the Basic Model 52

5 The Modified PageRank Algorithm: Dangling Nodes, Teleportation, and Convergence 53
5.1 The Problem of Dangling Pages 54
5.2 Repairing the Dangling Columns 55
5.3 The Random-Surfer Model with Teleportation 56
5.4 Derivation of the Google Matrix 57

5.5 Why Is the Google Matrix Strictly Positive and Stochastic? 58
5.6 Existence and Uniqueness of the PageRank Vector 60
5.7 A Numerical Example with a Dangling Page 61
5.8 Computing PageRank by Power Iteration 63
5.9 Why Teleportation Improves Convergence 64
5.10 An Efficient Form of the Iteration 66
5.11 Personalized PageRank 67
5.12 Mathematical Significance of the Modification 68

6 Numerical Implementation and Computational Results 69
6.1 Dataset Description 69
6.2 Construction of the Transition Matrix 70
6.3 Sparse PageRank Iteration 71
6.4 Effect of the Damping Factor 71
6.5 Numerical Verification 72
6.6 Ranking the Web Pages 73
6.7 Computational Discussion 74

7 Conclusion 75

1 Mathematical Preliminaries

Before establishing the Perron–Frobenius theorem, we introduce several definitions and elementary results concerning positive matrices, eigenvalues, and compact sets.

These ideas will play an essential role in the proof of the theorem and its subsequent applications to stochastic matrices and Markov chains.

1.1 Positive and Nonnegative Matrices

Definition 1.1 (Nonnegative Matrix)**.** A matrix $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ is called *nonnegative* if

$$a_{ij} \geq 0 \qquad \text{for all } i, j.$$

We write $A \geq 0$.

Definition 1.2 (Positive Matrix)**.** A matrix $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ is called *strictly positive* if

$$a_{ij} > 0 \qquad \text{for all } i, j.$$

We write $A > 0$.

For example, the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$$

is strictly positive, whereas

$$B = \begin{pmatrix} 2 & 0 \\ 3 & 4 \end{pmatrix}$$

is nonnegative but not strictly positive.

Throughout this note, inequalities involving vectors or matrices are understood *componentwise*.

Thus,

$$x \geq 0$$

means that every component of x is nonnegative, while

$$x > 0$$

means that every component is strictly positive.

Remark 1.3. The terminology *positive matrix* should not be confused with the notion of a *positive definite matrix*.

A real symmetric matrix A is positive definite if

$$x^T Ax > 0 \qquad \text{for every } x \neq 0.$$

This condition is fundamentally different from requiring every entry of A to be positive.

For example,

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

has strictly positive entries, but its eigenvalues are 3 and -1.

Consequently, A is not positive definite matrix, but it is clearly a positive matrix.

1.2 The Action of a Positive Matrix on Nonnegative Vectors

One of the most useful properties of strictly positive matrices concerns their action on nonnegative vectors.

Proposition 1.4. *Let $A \in \mathbb{R}^{n\times n}$ be strictly positive.*

If $x \in \mathbb{R}^n$ satisfies

$$x \geq 0, \qquad x \neq 0,$$

then

$$Ax > 0.$$

Proof. For each i,

$$(Ax)_i = \sum_{j=1}^{n} a_{ij}x_j.$$

Since $A > 0$, every coefficient a_{ij} is strictly positive.

Furthermore, $x \geq 0$ and $x \neq 0$, so at least one component, say x_k, satisfies $x_k > 0$.

Therefore,

$$(Ax)_i = \sum_{j=1}^{n} a_{ij}x_j \geq a_{ik}x_k > 0.$$

This holds for every i, and hence

$$Ax > 0.$$

□

This proposition is central to the Perron–Frobenius theorem.

It tells us that multiplying a nonzero, nonnegative vector by a strictly positive matrix produces a vector whose components are all strictly positive.

In particular, if a nonnegative eigenvector exists for a positive eigenvalue, this proposition will allow us to establish that every component of that eigenvector is strictly positive.

1.3 Eigenvalues and the Spectral Radius

Recall that a scalar $\lambda \in \mathbb{C}$ is an eigenvalue of a matrix $A \in \mathbb{R}^{n\times n}$ if there exists a nonzero vector $x \in \mathbb{C}^n$ such that

$$Ax = \lambda x.$$

The eigenvalues are precisely the roots of the characteristic polynomial

$$p_A(\lambda) = \det(\lambda I - A).$$

A real matrix may have complex eigenvalues. However, the eigenvalues of a finite square matrix form a finite set.

This motivates the following definition.

Definition 1.5 (Spectral Radius)**.** The *spectral radius* of a square matrix A is defined by

$$\rho(A) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } A\}. \tag{1}$$

Notice that the spectral radius is always a nonnegative real number.

For example, consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its characteristic polynomial is

$$\det(\lambda I - A) = (\lambda - 2)^2 - 1 = (\lambda - 3)(\lambda - 1).$$

Therefore, its eigenvalues are

$$\lambda_1 = 3, \qquad \lambda_2 = 1.$$

Hence,

$$\boxed{\rho(A) = 3.}$$

The Perron–Frobenius theorem will establish that when A is strictly positive, its spectral radius is itself a positive eigenvalue, with a strictly positive eigenvector.

Moreover, every other eigenvalue has strictly smaller absolute value.

1.4 Compactness and the Extreme Value Theorem

Our proof of the Perron–Frobenius theorem will involve an optimization problem.

To ensure that the optimal value is actually attained, we recall the notion of compactness.

Definition 1.6 (Compactness in Euclidean Space)**.** A subset $S \subseteq \mathbb{R}^n$ is called *compact* if it is closed and bounded.

This characterization follows from the Heine–Borel theorem.

A set is bounded if it is contained in a sufficiently large ball.

A set is closed if it contains the limits of all convergent sequences whose terms belong to the set.

For example, the interval $[0, 1]$ is compact, whereas $(0, 1)$ is not compact because it does not contain its endpoints.

Compactness is important because of the following classical result.

Theorem 1.7 (Extreme Value Theorem)**.** *Let $S \subseteq \mathbb{R}^n$ be nonempty and compact, and let*

$$f : S \longrightarrow \mathbb{R}$$

be continuous.

Then there exist points $x_{\min}, x_{\max} \in S$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max})$$

for every $x \in S$.

In other words, a continuous function defined on a nonempty compact set attains both its minimum and maximum.

1.5 The Standard Simplex

We now introduce a compact set that will be particularly useful in our proof.

Define

$$S = \left\{ x \in \mathbb{R}^n : x_i \geq 0, \quad \sum_{i=1}^{n} x_i = 1 \right\}. \tag{2}$$

This set is called the *standard simplex*.

When $n = 2$, it is the line segment joining $(1, 0)$ and $(0, 1)$.

When $n = 3$, it is the filled triangle whose vertices are

$$(1, 0, 0), \qquad (0, 1, 0), \qquad (0, 0, 1).$$

Proposition 1.8. *The standard simplex*

$$S = \left\{ x \in \mathbb{R}^n : x_i \geq 0, \quad \sum_{i=1}^{n} x_i = 1 \right\}$$

is nonempty and compact.

Proof. Recall that, by the Heine–Borel theorem, a subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

We will establish these properties separately.

Step 1: The set is nonempty. Consider the vector

$$e_1 = (1, 0, \ldots, 0)^T.$$

Every component of e_1 is nonnegative, and the sum of its components is one.

Therefore,

$$e_1 \in S.$$

Hence S is nonempty.

Step 2: The set is bounded. Let $x \in S$ be arbitrary.

By the definition of S,

$$x_i \geq 0, \qquad \sum_{i=1}^{n} x_i = 1.$$

Because all components are nonnegative and their sum is one, no individual component can exceed one.

Thus,

$$0 \leq x_i \leq 1, \qquad i = 1, \ldots, n.$$

Consequently,

$$S \subseteq [0, 1]^n.$$

Here, $[0, 1]^n$ denotes the set of all vectors in $\mathbb{R}^n$ whose components belong to $[0, 1]$.

In particular,

$$\|x\|_2^2 = \sum_{i=1}^{n} x_i^2 \leq n.$$

Therefore,

$$\|x\|_2 \leq \sqrt{n}$$

for every $x \in S$.

Hence S is bounded.

Step 3: Understanding sequences of vectors. Before proving that S is closed, let us recall how convergence of sequences extends from $\mathbb{R}$ to $\mathbb{R}^n$.

In elementary calculus, we study sequences of real numbers such as

$$a_1, a_2, a_3, \ldots$$

and write

$$a_k \longrightarrow a \qquad \text{as } k \to \infty$$

when the terms approach a real number a.

In $\mathbb{R}^n$, the idea is essentially the same, except that each term of the sequence is now a vector rather than a scalar.

We write such a sequence as

$$x^{(1)}, x^{(2)}, x^{(3)}, \ldots$$

The notation $x^{(k)}$ means the k-th vector in the sequence.

The superscript (k) is an index, not an exponent.

For example, in $\mathbb{R}^2$, consider the sequence

$$x^{(k)} = \begin{pmatrix} \frac{1}{k} \\ 1 - \frac{1}{k} \end{pmatrix}, \qquad k = 1, 2, 3, \ldots$$

Its first few terms are

$$x^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad x^{(2)} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \qquad x^{(3)} = \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \end{pmatrix}.$$

As k increases, the first component approaches zero, while the second approaches one.

Therefore,

$$x^{(k)} \longrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Notice that we have simply taken the limit of each component.

More generally, if

$$x^{(k)} = \begin{pmatrix} x_1^{(k)} \\ x_2^{(k)} \\ \vdots \\ x_n^{(k)} \end{pmatrix},$$

then

$$x^{(k)} \longrightarrow x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

if and only if

$$x_i^{(k)} \longrightarrow x_i \qquad \text{for every } i = 1, \ldots, n.$$

Thus, convergence in $\mathbb{R}^n$ is equivalent to convergence of each coordinate.

We now use this observation to establish closedness.

Step 4: The set is closed. Recall that a subset of $\mathbb{R}^n$ is closed if the limit of every convergent sequence of points in the set also belongs to the set.

Therefore, let

$$x^{(1)}, x^{(2)}, x^{(3)}, \ldots$$

be an arbitrary sequence of vectors belonging to S.

Suppose this sequence converges to some vector $x \in \mathbb{R}^n$:

$$x^{(k)} \longrightarrow x.$$

Our task is to show that x also belongs to S.

Since every $x^{(k)}$ belongs to S, each vector satisfies two conditions:

$$x_i^{(k)} \geq 0 \qquad \text{for every } i,$$

and

$$\sum_{i=1}^{n} x_i^{(k)} = 1.$$

We will show that both conditions remain true for the limiting vector x.

First condition: Nonnegative components.

Fix any coordinate i.

Since

$$x_i^{(k)} \geq 0$$

for every k, and

$$x_i^{(k)} \longrightarrow x_i,$$

the limit cannot be negative.

Indeed, a convergent sequence of nonnegative real numbers must have a nonnegative limit.

Therefore,

$$x_i \geq 0.$$

Since this argument applies to every coordinate,

$$\boxed{x \geq 0.}$$

Second condition: The components sum to one.

For every positive integer k, we know that

$$x_1^{(k)} + x_2^{(k)} + \cdots + x_n^{(k)} = 1.$$

Taking the limit as $k \to \infty$, we obtain

$$\lim_{k\to\infty} \left(x_1^{(k)} + \cdots + x_n^{(k)} \right) = 1.$$

Because the sum contains only finitely many terms, we may interchange the limit and the summation:

$$\sum_{i=1}^{n} \lim_{k\to\infty} x_i^{(k)} = 1.$$

But each coordinate converges:

$$\lim_{k\to\infty} x_i^{(k)} = x_i.$$

Consequently,

$$\boxed{\sum_{i=1}^{n} x_i = 1.}$$

We have therefore shown that the limiting vector x satisfies both conditions required for membership in S:

$$x_i \geq 0, \qquad \sum_{i=1}^{n} x_i = 1.$$

Hence,

$$x \in S.$$

Since the sequence was arbitrary, every convergent sequence of vectors in S has its limit in S.

Therefore, S is closed.

Step 5: Apply the Heine–Borel theorem. We have established that

1. S is nonempty;
2. S is bounded;
3. S is closed.

Since $S \subseteq \mathbb{R}^n$ is closed and bounded, the Heine–Borel theorem implies that S is compact.

Thus,

$$\boxed{S \text{ is nonempty and compact.}}$$

□

The importance of the simplex comes from the following observation.

Suppose $x \geq 0$, $x \neq 0$, and

$$Ax \geq tx.$$

Because

$$\sum_{i=1}^{n} x_i > 0,$$

we can define

$$y = \frac{x}{\sum_{i=1}^{n} x_i}.$$

Then $y \in S$ and

$$Ay \geq ty.$$

Thus, normalizing a nonzero nonnegative vector does not change the scalar t appearing in the inequality.

Consequently, when searching for the largest value of t such that

$$Ax \geq tx,$$

we may restrict our attention to vectors belonging to the compact set S.

This normalization will be the starting point of our proof of the Perron–Frobenius theorem.

2 The Perron–Frobenius Theorem

We now establish one of the fundamental results in the theory of nonnegative matrices.

The Perron–Frobenius theorem reveals that the eigenvalue structure of a strictly positive matrix is considerably more restrictive than that of an arbitrary real matrix.

In particular, positivity guarantees the existence of a distinguished positive eigenvalue and a corresponding eigenvector whose components are all strictly positive.

We begin with the theorem and then develop its proof in several stages.

2.1 Statement of the Theorem

Theorem 2.1 (Perron–Frobenius Theorem for Strictly Positive Matrices)**.** *Let $A \in \mathbb{R}^{n\times n}$ be a strictly positive matrix, meaning that*

$$a_{ij} > 0 \qquad \text{for every } i, j.$$

Then the following statements hold.

(i) The spectral radius $\rho(A)$ is a strictly positive eigenvalue of A.

(ii) There exists a corresponding eigenvector $x > 0$ such that

$$Ax = \rho(A)x.$$

(iii) Every other eigenvalue $\mu \neq \rho(A)$ satisfies

$$|\mu| < \rho(A).$$

(iv) The eigenvalue $\rho(A)$ is algebraically simple.

The eigenvalue $\rho(A)$ is called the *Perron eigenvalue*, and its corresponding strictly positive eigenvector is called a *Perron eigenvector*.

The third assertion means that the Perron eigenvalue is strictly dominant: no other eigenvalue has the same absolute value.

We now establish these assertions one at a time.

2.2 Existence of a Positive Eigenvalue and Eigenvector

Our first objective is to prove the existence of a positive scalar λ and a nonzero, nonnegative vector x satisfying

$$Ax = \lambda x.$$

The challenge is that we cannot assume such an eigenvector exists.

Instead, we begin with a weaker condition:

$$Ax \geq tx, \qquad x \geq 0, \quad x \neq 0. \tag{3}$$

The inequalities are understood componentwise.

This condition is easier to satisfy than the eigenvalue equation.

Indeed, since $A > 0$, every nonzero, nonnegative vector satisfies

$$Ax > 0.$$

Consequently, $t = 0$ is always an admissible choice.

Our strategy is to determine the largest scalar t for which the inequality can hold and then prove that the inequality must become an equality at this largest value.

2.2.1 Normalization of the Vector

Recall the standard simplex

$$S = \left\{ x \in \mathbb{R}^n : x \geq 0, \quad \sum_{i=1}^{n} x_i = 1 \right\}. \tag{4}$$

We have already established that S is nonempty and compact.

Suppose that a nonzero, nonnegative vector x satisfies

$$Ax \geq tx.$$

Define

$$y = \frac{x}{\sum_{i=1}^{n} x_i}.$$

Since the denominator is positive, we obtain

$$y \in S$$

and

$$Ay \geq ty.$$

Thus, normalization preserves the admissible scalar t.

We may therefore restrict our search to the compact set S without excluding any possible value of t.

2.2.2 Existence of the Maximum

We have established that searching for a scalar t and a nonzero, nonnegative vector x satisfying

$$Ax \geq tx$$

is equivalent to restricting our attention to vectors in the standard simplex

$$S = \left\{ x \in \mathbb{R}^n : x_i \geq 0, \quad \sum_{i=1}^{n} x_i = 1 \right\}.$$

Our objective is to find the largest possible scalar t for which this inequality can hold.

However, before speaking of a largest value, we must answer two questions:

1. Does there exist at least one admissible scalar t?
2. Can the admissible scalars become arbitrarily large, or is there a finite upper bound?

Once these questions are answered, we will investigate whether the largest possible value is actually attained.

Step 1: Defining the admissible scalars. We define

$$\mathcal{T} = \{t \geq 0 : \text{there exists } x \in S \text{ such that } Ax \geq tx\}.$$

Thus, a scalar t belongs to $\mathcal{T}$ if we can find at least one normalized, nonnegative vector x for which

$$Ax \geq tx.$$

Notice that we do not require the inequality to hold for every vector in S. Finding just one suitable vector is sufficient.

We first observe that $\mathcal{T}$ is nonempty.

Indeed, because $A > 0$, every $x \in S$ satisfies

$$Ax > 0.$$

Consequently,

$$Ax \geq 0 = 0x.$$

Thus, $t = 0$ is admissible, and hence

$$0 \in \mathcal{T}.$$

We conclude that

$$\boxed{\mathcal{T} \neq \varnothing.}$$

Step 2: Establishing an upper bound. We now show that the admissible scalars cannot grow without bound.

Suppose that $t \in \mathcal{T}$.

By definition, there exists a vector $x \in S$ satisfying

$$Ax \geq tx. \tag{5}$$

Since this inequality holds componentwise, we have

$$(Ax)_i \geq tx_i, \qquad i = 1, \ldots, n.$$

Adding these n inequalities gives

$$\sum_{i=1}^{n} (Ax)_i \geq t \sum_{i=1}^{n} x_i.$$

But $x \in S$, so

$$\sum_{i=1}^{n} x_i = 1.$$

Therefore,

$$t \leq \sum_{i=1}^{n} (Ax)_i. \tag{6}$$

This is already useful.

It tells us that any admissible scalar t cannot exceed the sum of the components of Ax.

However, the right-hand side still depends on our choice of x.

We therefore seek an intrinsic upper bound that depends only on the matrix A.

Recall that

$$(Ax)_i = \sum_{j=1}^{n} a_{ij} x_j.$$

Consequently,

$$\begin{aligned} \sum_{i=1}^{n} (Ax)_i &= \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_j \\ &= \sum_{j=1}^{n} \left(\sum_{i=1}^{n} a_{ij} \right) x_j. \end{aligned}$$

The second equality follows by rearranging the order of two finite sums.

Notice that

$$\sum_{i=1}^{n} a_{ij}$$

is the sum of the entries in the j-th column of A.

Let M denote the largest of these column sums:

$$M = \max_{1 \leq j \leq n} \sum_{i=1}^{n} a_{ij}. \tag{7}$$

By the definition of M,

$$\sum_{i=1}^{n} a_{ij} \leq M$$

for every j.

Since $x_j \geq 0$, we may multiply both sides by x_j without reversing the inequality:

$$\left(\sum_{i=1}^{n} a_{ij}\right) x_j \leq M x_j.$$

Adding over j gives

$$\begin{aligned}\sum_{j=1}^{n} \left(\sum_{i=1}^{n} a_{ij}\right) x_j &\leq \sum_{j=1}^{n} M x_j \\ &= M \sum_{j=1}^{n} x_j \\ &= M.\end{aligned}$$

Combining this with (6), we obtain

$$\boxed{0 \leq t \leq M \qquad \text{for every } t \in \mathcal{T}.} \tag{8}$$

This establishes that $\mathcal{T}$ is bounded above.

Most importantly, M depends only on the entries of A, not on the particular vector x.

Step 3: Introducing the supremum. We have now established that $\mathcal{T}$ is nonempty and bounded above.

By the completeness property of the real numbers, every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound, or supremum.

We may therefore define

$$\boxed{t_{\max} = \sup \mathcal{T}.} \tag{9}$$

Because every admissible scalar lies between 0 and M, we know that

$$0 \leq t_{\max} \leq M.$$

At this stage, however, we must be careful.

The supremum of a set need not belong to the set itself.

For example, the interval

$$(0, 1)$$

has supremum 1, but 1 does not belong to the interval.

Therefore, the existence of $t_{\max}$ does not yet guarantee that there is a vector $x \in S$ satisfying

$$Ax \geq t_{\max} x.$$

This is precisely what we must establish next.

Step 4: Constructing a maximizing sequence. Since $t_{\max}$ is the supremum of $\mathcal{T}$, we can find admissible scalars arbitrarily close to $t_{\max}$.

More precisely, for every positive integer k, there exists $t_k \in \mathcal{T}$ such that

$$t_{\max} - \frac{1}{k} < t_k \leq t_{\max}.$$

Consequently,

$$t_k \longrightarrow t_{\max} \qquad \text{as } k \to \infty. \tag{10}$$

For each $t_k \in \mathcal{T}$, the definition of admissibility guarantees the existence of a vector $x^{(k)} \in S$ satisfying

$$Ax^{(k)} \geq t_k x^{(k)}. \tag{11}$$

We have therefore constructed two sequences:

$$t_1, t_2, t_3, \ldots$$

and

$$x^{(1)}, x^{(2)}, x^{(3)}, \ldots$$

The scalars t_k approach $t_{\max}$.

Our remaining task is to find a suitable limiting vector.

Step 5: Applying compactness. We have already proved that the standard simplex S is compact.

One consequence of compactness in $\mathbb{R}^n$ is that every sequence in S possesses a convergent subsequence whose limit also belongs to S.

Therefore, the sequence $(x^{(k)})$ has a subsequence $(x^{(k_j)})$ such that

$$x^{(k_j)} \longrightarrow x^* \qquad \text{for some } x^* \in S. \tag{12}$$

This is the precise point where compactness enters our argument.

Without compactness, the candidate vectors might approach a limiting vector outside the admissible set, or might fail to possess a convergent subsequence.

Because $t_k \to t_{\max}$, the corresponding subsequence also satisfies

$$t_{k_j} \longrightarrow t_{\max}.$$

For every j, we have

$$Ax^{(k_j)} \geq t_{k_j} x^{(k_j)}.$$

We now pass to the limit.

Since A is fixed, matrix multiplication is continuous:

$$Ax^{(k_j)} \longrightarrow Ax^*.$$

Similarly, scalar multiplication is continuous, so

$$t_{k_j} x^{(k_j)} \longrightarrow t_{\max} x^*.$$

Because the inequalities hold componentwise, we may take the limit in each component.

Thus,

$$\boxed{Ax^* \geq t_{\max} x^*.} \tag{13}$$

Furthermore, since $x^* \in S$,

$$x^* \geq 0, \qquad \sum_{i=1}^{n} x_i^* = 1.$$

In particular,

$$x^* \neq 0.$$

Therefore, $t_{\max}$ is admissible:

$$t_{\max} \in \mathcal{T}.$$

We have proved that the supremum is actually attained.

Hence,

$$\boxed{t_{\max} = \max \mathcal{T}.} \tag{14}$$

There exists a nonzero, nonnegative vector x^* satisfying

$$Ax^* \geq t_{\max} x^*,$$

and no larger admissible scalar can exist.

This completes the existence argument.

2.2.3 Why the Inequality Must Become an Equality

We now establish the crucial step of the proof.

We claim that

$$Ax^* = t_{\max} x^*. \tag{15}$$

Suppose, for contradiction, that equality does not hold.

Since

$$Ax^* \geq t_{\max} x^*,$$

the vector

$$y = Ax^* - t_{\max} x^*$$

satisfies

$$y \geq 0, \qquad y \neq 0.$$

Because A is strictly positive, our earlier proposition implies

$$Ay > 0.$$

Substituting the definition of y, we obtain

$$A(Ax^* - t_{\max}x^*) > 0.$$

By linearity,

$$A^2x^* - t_{\max}Ax^* > 0.$$

Hence,

$$A^2x^* > t_{\max}Ax^*. \tag{16}$$

Now define

$$z = Ax^*.$$

Since $A > 0$ and $x^* \geq 0$, $x^* \neq 0$,

$$z > 0.$$

Equation (16) becomes

$$Az > t_{\max}z.$$

Therefore, for every component,

$$(Az)_i > t_{\max}z_i.$$

Since $z_i > 0$, division gives

$$\frac{(Az)_i}{z_i} > t_{\max}.$$

There are only finitely many components, so we may define

$$t_{\text{new}} = \min_{1\leq i\leq n} \frac{(Az)_i}{z_i}.$$

Every ratio exceeds $t_{\max}$, and hence

$$t_{\text{new}} > t_{\max}.$$

By the definition of the minimum,

$$Az \geq t_{\text{new}}z.$$

Since $z > 0$, we may normalize z to obtain a vector in S without changing this inequality. Thus, t_{new} is an admissible scalar.

But

$$t_{\text{new}} > t_{\max},$$

contradicting the maximality of $t_{\max}$.

Therefore, our assumption was false, and we conclude that

$$\boxed{Ax^* = t_{\max}x^*.} \tag{17}$$

Thus, $t_{\max}$ is an eigenvalue of A, and x^* is a corresponding nonnegative eigenvector.

2.2.4 Strict Positivity of the Eigenvector

We have established that

$$Ax^* = t_{\max}x^*, \qquad x^* \geq 0, \quad x^* \neq 0.$$

Because $A > 0$, we know that

$$Ax^* > 0.$$

Furthermore, since $t_{\max} \geq 0$, the equality

$$t_{\max}x^* = Ax^* > 0$$

implies that

$$t_{\max} > 0.$$

It follows that

$$x^* > 0.$$

We have therefore established the existence of a strictly positive eigenvalue and a corresponding strictly positive eigenvector.

2.3 The Spectral Radius and Dominance

Our next objective is to prove that the positive eigenvalue $t_{\max}$ dominates every other eigenvalue in absolute value.

The argument rests on a simple inequality involving complex vectors.

Lemma 2.2. *Let $A \in \mathbb{R}^{n\times n}$ be a nonnegative matrix.*

Then, for every $v \in \mathbb{C}^n$,

$$|Av| \leq A|v|, \tag{18}$$

where absolute values are taken componentwise.

Proof. For each component i,

$$(Av)_i = \sum_{j=1}^{n} a_{ij} v_j.$$

By the triangle inequality,

$$\left| \sum_{j=1}^{n} a_{ij} v_j \right| \leq \sum_{j=1}^{n} |a_{ij} v_j|.$$

Since $a_{ij} \geq 0$,

$$|a_{ij} v_j| = a_{ij} |v_j|.$$

Therefore,

$$|(Av)_i| \leq \sum_{j=1}^{n} a_{ij} |v_j| = (A|v|)_i.$$

Since this holds for every i,

$$|Av| \leq A|v|.$$

□

We now apply this lemma.

Let μ be any eigenvalue of A, and let $v \neq 0$ be a corresponding eigenvector.

Thus,

$$Av = \mu v.$$

Taking componentwise absolute values gives

$$|Av| = |\mu||v|.$$

By Lemma 2.2,

$$|Av| \leq A|v|.$$

Combining these relations,

$$A|v| \geq |\mu||v|. \tag{19}$$

Notice that

$$|v| \geq 0, \qquad |v| \neq 0.$$

Consequently, the pair

$$t = |\mu|, \qquad x = |v|$$

satisfies the same inequality used to define $t_{\max}$.

By the maximality of $t_{\max}$,

$$|\mu| \leq t_{\max}.$$

Since $t_{\max}$ is itself an eigenvalue,

$$\boxed{\rho(A) = t_{\max}.} \tag{20}$$

Thus, the spectral radius of a strictly positive matrix is a positive eigenvalue with a strictly positive eigenvector.

2.4 Strict Dominance of the Perron Eigenvalue

In the preceding subsection, we established that the spectral radius $\rho(A)$ is a strictly positive eigenvalue of A, with a corresponding strictly positive eigenvector.

We also proved that every eigenvalue μ of A satisfies

$$|\mu| \leq \rho(A).$$

However, this inequality alone does not establish that $\rho(A)$ is the unique eigenvalue of maximum modulus.

There might, in principle, exist another eigenvalue $\mu \neq \rho(A)$ satisfying

$$|\mu| = \rho(A).$$

For example, the two distinct eigenvalues 3 and -3 have the same absolute value.

Our objective is to rule out this possibility when every entry of A is strictly positive.

More precisely, we will prove that

$$\boxed{\mu \neq \rho(A) \quad \Longrightarrow \quad |\mu| < \rho(A).}$$

The key observation is that equality in the triangle inequality imposes strong restrictions on the arguments of complex numbers.

Step 1: Consider an eigenvalue of maximum modulus. Suppose μ is an eigenvalue of A satisfying

$$|\mu| = \rho(A). \tag{21}$$

Our goal is to establish that

$$\mu = \rho(A).$$

Let $v \in \mathbb{C}^n$ be a corresponding nonzero eigenvector:

$$Av = \mu v. \tag{22}$$

We allow v to have complex components, since the eigenvalues and eigenvectors of a real nonsymmetric matrix need not all be real.

Recall that $|v|$ denotes the vector obtained by taking the absolute value of each component:

$$|v| = \begin{pmatrix} |v_1| \\ \vdots \\ |v_n| \end{pmatrix}.$$

Since $v \neq 0$, we have

$$|v_i| \geq 0, \qquad |v| \neq 0.$$

Step 2: Show that $|v|$ is also an eigenvector. Recall that for a nonnegative matrix, the triangle inequality gives

$$|Av| \leq A|v|.$$

Since $Av = \mu v$, we obtain

$$|\mu||v| \leq A|v|.$$

Using $|\mu| = \rho(A)$, this becomes

$$A|v| \geq \rho(A)|v|. \tag{23}$$

Now recall an important result from the existence proof.

We defined $t_{\max}$ to be the largest scalar for which there exists a nonzero, nonnegative vector x satisfying

$$Ax \geq t_{\max} x.$$

We then proved two facts:

$$t_{\max} = \rho(A),$$

and, at this maximum, the inequality must become an equality.

In fact, the contradiction argument showed that *any* nonzero, nonnegative vector satisfying the inequality at $t_{\max}$ must satisfy equality.

Since $|v|$ is nonnegative and nonzero, equation (23) therefore implies

$$\boxed{A|v| = \rho(A)|v|.} \tag{24}$$

Thus, $|v|$ is itself an eigenvector corresponding to the Perron eigenvalue.

Furthermore, because $A > 0$ and $|v| \geq 0$, $|v| \neq 0$, we know that

$$A|v| > 0.$$

Since $\rho(A) > 0$, equation (24) implies

$$|v| > 0. \tag{25}$$

Consequently, every component v_j is nonzero.

This observation will be important when we examine equality in the triangle inequality.

Step 3: Obtain equality in the triangle inequality. Let us examine the eigenvalue equation component by component.

From (22), we have

$$(Av)_i = \mu v_i.$$

Expanding the matrix product,

$$\sum_{j=1}^{n} a_{ij} v_j = \mu v_i.$$

Taking absolute values gives

$$\left| \sum_{j=1}^{n} a_{ij} v_j \right| = |\mu| |v_i|. \tag{26}$$

Because $|\mu| = \rho(A)$, we obtain

$$\left| \sum_{j=1}^{n} a_{ij} v_j \right| = \rho(A) |v_i|.$$

On the other hand, equation (24) tells us that

$$A|v| = \rho(A)|v|.$$

Looking at its i-th component, we have

$$\sum_{j=1}^{n} a_{ij} |v_j| = \rho(A)|v_i|.$$

Combining the preceding two equalities gives

$$\boxed{\left| \sum_{j=1}^{n} a_{ij} v_j \right| = \sum_{j=1}^{n} a_{ij} |v_j|.} \tag{27}$$

Notice that this equality holds for every row index i.

Normally, the triangle inequality only guarantees

$$\left| \sum_{j=1}^{n} a_{ij} v_j \right| \leq \sum_{j=1}^{n} a_{ij} |v_j|.$$

But here we have equality.

We now investigate what this equality tells us about the components of v.

Step 4: Equality forces all components to have the same complex argument. Recall that every nonzero complex number can be written in polar form:

$$z = re^{i\theta}, \qquad r = |z| > 0.$$

The angle θ is called the argument of z.

Geometrically, the triangle inequality states that the length of the sum of several vectors in the complex plane cannot exceed the sum of their individual lengths.

For example, if

$$z_1 = 1, \qquad z_2 = i,$$

then

$$|z_1 + z_2| = |1 + i| = \sqrt{2},$$

whereas

$$|z_1| + |z_2| = 2.$$

Thus,

$$|z_1 + z_2| < |z_1| + |z_2|.$$

The inequality is strict because the two complex numbers point in different directions.

However, if

$$z_1 = 2, \qquad z_2 = 3,$$

then both numbers point in the same direction along the real axis, and

$$|z_1 + z_2| = 5 = |z_1| + |z_2|.$$

More generally, for nonzero complex numbers $z_1, \ldots, z_n$,

$$\left| \sum_{j=1}^{n} z_j \right| = \sum_{j=1}^{n} |z_j| \tag{28}$$

holds if and only if all the numbers z_j have the same complex argument.

Equivalently, there exists a real number θ such that

$$z_j = e^{i\theta}|z_j|, \qquad j = 1, \ldots, n.$$

We apply this observation to (27).

Fix any row index i and define

$$z_j = a_{ij} v_j.$$

We have already established that $v_j \neq 0$ for every j.

Furthermore, since $A > 0$,

$$a_{ij} > 0.$$

Therefore, every z_j is nonzero.

Equation (27) becomes

$$\left| \sum_{j=1}^{n} z_j \right| = \sum_{j=1}^{n} |z_j|.$$

Hence all the numbers

$$a_{i1}v_1,\ a_{i2}v_2,\ \ldots,\ a_{in}v_n$$

must have the same complex argument.

But multiplying a complex number by a strictly positive real scalar does not change its argument.

Therefore,

$$\arg(a_{ij}v_j) = \arg(v_j).$$

Consequently, the components

$$v_1, v_2, \ldots, v_n$$

all have the same complex argument.

Thus, there exists a real number θ such that

$$\boxed{v_j = e^{i\theta}|v_j|, \qquad j = 1, \ldots, n.} \tag{29}$$

In vector notation,

$$\boxed{v = e^{i\theta}|v|.} \tag{30}$$

This is the crucial consequence of equality in the triangle inequality.

Although v may initially have complex components, all those components point in the same direction in the complex plane.

Therefore, v differs from the positive vector $|v|$ only by a common complex factor $e^{i\theta}$.

Step 5: Conclude that $\mu = \rho(A)$. We now return to the eigenvalue equation

$$Av = \mu v.$$

Substituting (30), we obtain

$$A(e^{i\theta}|v|) = \mu e^{i\theta}|v|.$$

Since $e^{i\theta}$ is a nonzero scalar, linearity gives

$$e^{i\theta}A|v| = \mu e^{i\theta}|v|.$$

Canceling the common factor, we obtain

$$A|v| = \mu|v|. \tag{31}$$

But in Step 2, we already proved

$$A|v| = \rho(A)|v|.$$

Comparing these two equations,

$$\mu|v| = \rho(A)|v|.$$

Therefore,

$$(\mu - \rho(A))|v| = 0.$$

Since $|v| \neq 0$, the scalar multiplying it must be zero.

Hence,

$$\boxed{\mu = \rho(A).}$$

We have therefore proved that any eigenvalue whose modulus equals $\rho(A)$ must actually be $\rho(A)$.

Consequently, every distinct eigenvalue satisfies

$$\boxed{|\mu| < \rho(A), \qquad \mu \neq \rho(A).} \tag{32}$$

This establishes the strict dominance of the Perron eigenvalue.

2.5 Algebraic Simplicity

We have proved that $\rho(A)$ is the only eigenvalue of A whose modulus equals the spectral radius.

However, this does not yet establish that $\rho(A)$ occurs only once as a root of the characteristic polynomial.

An eigenvalue may be the only eigenvalue of maximum modulus and still have algebraic multiplicity greater than one.

Therefore, to complete the Perron–Frobenius theorem, we must prove that $\rho(A)$ is algebraically simple.

We divide the argument into two parts.

First, we establish that the eigenspace corresponding to $\rho(A)$ is one-dimensional.

We then rule out the existence of generalized eigenvectors associated with this eigenvalue.

2.5.1 Geometric Multiplicity

Let

$$\lambda = \rho(A)$$

and choose a strictly positive eigenvector x:

$$Ax = \lambda x, \qquad x > 0.$$

Suppose there exists another real eigenvector y satisfying

$$Ay = \lambda y$$

that is linearly independent of x.

Since every component of x is strictly positive, the ratios y_i/x_i are well-defined.

Set

$$c = \min_{1 \le i \le n} \frac{y_i}{x_i}$$

and define

$$z = y - cx.$$

By the definition of c,

$$z_i = y_i - cx_i \ge 0$$

for every i.

Furthermore, the minimum is attained for at least one index k, so

$$z_k = 0.$$

Because x and y are linearly independent,

$$z \neq 0.$$

Thus,

$$z \geq 0, \qquad z \neq 0, \qquad z_k = 0.$$

By linearity,

$$\begin{aligned} Az &= A(y - cx) \\ &= Ay - cAx \\ &= \lambda y - c\lambda x \\ &= \lambda z. \end{aligned}$$

But $A > 0$ and z is a nonzero, nonnegative vector.

Therefore,

$$Az > 0.$$

On the other hand,

$$(Az)_k = \lambda z_k = 0,$$

which is a contradiction.

Hence every real eigenvector corresponding to λ is a scalar multiple of x.

Since A and λ are real, the real and imaginary parts of any complex eigenvector are also real eigenvectors.

Therefore, the eigenspace over $\mathbb{C}$ is one-dimensional as well:

$$\boxed{\dim \ker(A - \lambda I) = 1.} \tag{33}$$

Thus, the Perron eigenvalue has geometric multiplicity one.

2.5.2 Ruling Out Generalized Eigenvectors

Geometric multiplicity one does not automatically imply algebraic multiplicity one.

For example, the matrix

$$B = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

has eigenvalue 2 with algebraic multiplicity two, but its corresponding eigenspace has dimension one.

We must therefore prove that no nontrivial Jordan chain exists for λ.

Suppose, for contradiction, that λ has algebraic multiplicity greater than one.

Because its eigenspace is one-dimensional, the theory of Jordan canonical form implies that there exists a real generalized eigenvector w satisfying

$$(A - \lambda I)w = x, \tag{34}$$

where $x > 0$ is our Perron eigenvector.

Equivalently,

$$Aw = \lambda w + x.$$

Applying A once more,

$$A^2 w = \lambda A w + Ax.$$

Since $Ax = \lambda x$,

$$A^2 w = \lambda^2 w + 2\lambda x.$$

Continuing inductively, we obtain

$$A^k w = \lambda^k w + k\lambda^{k-1} x. \tag{35}$$

Dividing by λ^k gives

$$\frac{A^k w}{\lambda^k} = w + \frac{k}{\lambda} x. \tag{36}$$

Since $\lambda > 0$ and $x > 0$, the right-hand side is unbounded as $k \to \infty$.

We now show that the left-hand side must remain bounded.

Because every component x_i is strictly positive, we may choose

$$C = \max_{1 \le i \le n} \frac{|w_i|}{x_i}.$$

Then

$$|w| \le Cx$$

componentwise.

Since A is nonnegative,

$$|A^k w| \le A^k |w|.$$

Using $|w| \le Cx$,

$$A^k |w| \le C A^k x.$$

But

$$A^k x = \lambda^k x.$$

Therefore,

$$|A^k w| \leq C\lambda^k x.$$

Since $\lambda > 0$,

$$\boxed{\left|\frac{A^k w}{\lambda^k}\right| \leq Cx.} \tag{37}$$

The right-hand side is independent of k.

Hence the sequence

$$\left(\frac{A^k w}{\lambda^k}\right)_{k=1}^{\infty}$$

is bounded.

This contradicts (36), whose right-hand side grows without bound.

Therefore, no such generalized eigenvector exists.

We conclude that

$$\boxed{\rho(A) \text{ is algebraically simple.}} \tag{38}$$

2.6 Summary of the Proof

We have established all four assertions of the Perron–Frobenius theorem for strictly positive matrices.

The essential ingredients were the following:

1. Compactness of the standard simplex allowed us to establish the existence of a maximum scalar $t_{\max}$ satisfying

$$Ax \geq t_{\max} x.$$

2. Strict positivity of A forced the maximizing inequality to become an equality:

$$Ax = t_{\max} x.$$

3. The componentwise inequality

$$|Av| \leq A|v|$$

established that

$$t_{\max} = \rho(A).$$

4. Equality in the complex triangle inequality established strict dominance of the Perron eigenvalue.

5. Positivity and a boundedness argument ruled out additional eigenvectors and generalized eigenvectors corresponding to the Perron eigenvalue.

Consequently, every strictly positive matrix possesses a unique dominant eigenvalue

$$\lambda_1 = \rho(A) > 0,$$

which is algebraically simple and has a strictly positive eigenvector.

Since its eigenspace is one-dimensional, the Perron eigenvector is unique up to multiplication by a nonzero scalar.

In particular, there is exactly one Perron eigenvector whose components are positive and sum to one.

3 Stochastic Matrices and Stationary Probability Distributions

In the preceding section, we established the Perron–Frobenius theorem for strictly positive matrices.

We proved that such a matrix possesses a distinguished positive eigenvalue, equal to its spectral radius, together with a strictly positive eigenvector.

These results become particularly important when we study matrices whose entries represent probabilities.

Such matrices arise naturally in Markov chains, population models, random walks, and web-page ranking algorithms.

We begin by introducing stochastic matrices and explaining their relationship to the eigenvalue problem.

3.1 What Is a Stochastic Matrix?

Suppose a system can occupy one of n possible states.

At each step, the system may remain in its current state or move to another state.

We would like to represent the probabilities of these transitions using a matrix.

For example, suppose a system has only two states.

If the system is currently in state 1, suppose it has an 80% probability of remaining there and a 20% probability of moving to state 2.

Similarly, if it is in state 2, suppose it has a 30% probability of moving to state 1 and a 70% probability of remaining in state 2.

We may represent these probabilities by the matrix

$$P = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}. \tag{39}$$

Here, the entry p_{ij} represents the probability of moving *from state j to state i*.

For example,

$$p_{21} = 0.2$$

represents the probability of moving from state 1 to state 2.

Notice that the entries in each column sum to one:

$$0.8 + 0.2 = 1, \qquad 0.3 + 0.7 = 1.$$

This is necessary because the total probability of all possible destinations from a given state must equal one.

We are therefore led to the following definition.

Definition 3.1 (Column-Stochastic Matrix)**.** A matrix $P = (p_{ij}) \in \mathbb{R}^{n\times n}$ is called *column-stochastic* if

$$p_{ij} \geq 0 \qquad \text{for all } i, j,$$

and

$$\sum_{i=1}^{n} p_{ij} = 1, \qquad j = 1, \ldots, n. \tag{40}$$

Thus, a column-stochastic matrix is a nonnegative matrix whose columns each sum to one.

Some textbooks instead use row-stochastic matrices, whose rows sum to one.

Both conventions are valid.

Throughout this note, we use the column-stochastic convention because probability distributions will be represented by column vectors.

Remark 3.2. A column-stochastic matrix need not have strictly positive entries.

For example,

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is column-stochastic, even though some entries are zero.

We call a column-stochastic matrix *strictly positive* if every entry is positive.

This distinction will become important when we discuss uniqueness and convergence.

3.2 Probability Distributions as Vectors

Suppose our system has n possible states.

Let x_i denote the probability that the system is currently in state i.

The probability distribution can be represented by the vector

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Since probabilities cannot be negative,

$$x_i \geq 0$$

for every i.

Furthermore, the system must occupy one of the n states, so

$$\sum_{i=1}^{n} x_i = 1.$$

Thus,

$$\boxed{x \geq 0, \qquad \sum_{i=1}^{n} x_i = 1.}$$

Notice something familiar!

These are precisely the conditions defining the standard simplex introduced in Section 2.

Therefore, every probability distribution over n states can be represented as a vector belonging to the standard simplex.

This provides a direct connection between our earlier compactness argument and probability theory.

3.3 How a Stochastic Matrix Updates a Probability Distribution

Suppose P is column-stochastic and $x^{(0)}$ represents the initial probability distribution.

The superscript (0) indicates the initial time step; it is not an exponent.

After one transition, the probability distribution becomes

$$x^{(1)} = Px^{(0)}. \tag{41}$$

After two transitions,

$$x^{(2)} = Px^{(1)} = P^2x^{(0)}.$$

Continuing in this manner, we obtain

$$\boxed{x^{(k)} = P^k x^{(0)}.} \tag{42}$$

This describes a discrete-time Markov chain with a fixed transition matrix P.

But an important question arises:

Does multiplication by P preserve the properties of a probability distribution?

We answer this in the following proposition.

Proposition 3.3. *Let P be a column-stochastic matrix and let x be a probability distribution. Then Px is also a probability distribution.*

Proof. Since $P \geq 0$ and $x \geq 0$,

$$(Px)_i = \sum_{j=1}^{n} p_{ij} x_j \geq 0.$$

Therefore,

$$Px \geq 0.$$

Next, we must show that the components of Px sum to one.

Let

$$\mathbf{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Because the columns of P sum to one,

$$\mathbf{1}^T P = \mathbf{1}^T. \tag{43}$$

Therefore,

$$\begin{aligned} \mathbf{1}^T (Px) &= (\mathbf{1}^T P)x \\ &= \mathbf{1}^T x \\ &= 1. \end{aligned}$$

Consequently, Px has nonnegative components whose sum is one.

Hence Px is a probability distribution.

□

This proposition tells us that a stochastic matrix maps the standard simplex into itself.

Starting from a valid probability distribution, repeated multiplication by P will continue to produce valid probability distributions.

3.4 Why Is the Eigenvalue One Special?

Recall that the eigenvalue problem for a matrix P takes the form

$$Px = \lambda x.$$

For a column-stochastic matrix, the number 1 plays a distinguished role.

We first establish that it is always an eigenvalue.

Proposition 3.4. *Let $P \in \mathbb{R}^{n\times n}$ be a column-stochastic matrix.*

Then 1 is an eigenvalue of P, and its spectral radius satisfies

$$\boxed{\rho(P) = 1.}$$

Proof. Recall that P is column-stochastic, meaning that every entry is nonnegative and the entries in each column sum to one.

Let

$$\mathbf{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^n.$$

We divide the proof into three steps.

Step 1: Show that 1 is an eigenvalue of P. Since every column of P sums to one, we have

$$\mathbf{1}^T P = \mathbf{1}^T.$$

Taking the transpose of both sides gives

$$P^T \mathbf{1} = \mathbf{1}.$$

Equivalently,

$$P^T \mathbf{1} = 1 \cdot \mathbf{1}.$$

Since $\mathbf{1} \neq 0$, this is precisely the eigenvalue equation for P^T, with eigenvalue 1.

Therefore, 1 is an eigenvalue of P^T.

However, our objective is to establish that 1 is an eigenvalue of P itself.

Recall that a matrix and its transpose have the same characteristic polynomial.

Indeed,

$$\begin{aligned} \det(\lambda I - P^T) &= \det((\lambda I - P)^T) \\ &= \det(\lambda I - P), \end{aligned}$$

where we used the identity

$$\det(B^T) = \det(B).$$

Thus, P and P^T have the same characteristic polynomial, and consequently the same eigenvalues.
Since 1 is an eigenvalue of P^T, it must also be an eigenvalue of P.
We have therefore established that

$$\boxed{1 \text{ is an eigenvalue of } P.}$$

Step 2: Show that $\rho(P) \geq 1$. Recall the definition of the spectral radius of a square matrix:

$$\rho(P) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } P\}.$$

In other words, the spectral radius is the largest absolute value among all the eigenvalues of P.
Suppose the eigenvalues of P are

$$\lambda_1, \lambda_2, \ldots, \lambda_n,$$

listed according to their algebraic multiplicities.
Then

$$\rho(P) = \max\{|\lambda_1|, |\lambda_2|, \ldots, |\lambda_n|\}.$$

From Step 1, we know that at least one of these eigenvalues is 1.
Therefore, the set of absolute values contains the number

$$|1| = 1.$$

Since the maximum of a set cannot be smaller than one of its elements, we obtain

$$\boxed{\rho(P) \geq 1.}$$

Notice that this inequality does not yet establish $\rho(P) = 1$.
There might, in principle, exist another eigenvalue whose absolute value is greater than one.
We must now rule out that possibility.

Step 3: Show that $\rho(P) \leq 1$. Let λ be an arbitrary eigenvalue of P.
Since P is a real matrix, its eigenvalues may be complex.
Therefore, let $v \in \mathbb{C}^n$ be a corresponding nonzero eigenvector:

$$Pv = \lambda v.$$

Our objective is to prove that

$$|\lambda| \leq 1.$$

Recall that for a nonnegative matrix, the triangle inequality gives

$$|Pv| \leq P|v|,$$

where absolute values are taken componentwise.

Since $Pv = \lambda v$,

$$|Pv| = |\lambda||v|.$$

Combining these relations,

$$|\lambda||v| \leq P|v|. \tag{44}$$

Let us examine this inequality component by component.

For every i,

$$|\lambda||v_i| \leq \sum_{j=1}^{n} p_{ij}|v_j|.$$

Adding these inequalities over all i gives

$$|\lambda| \sum_{i=1}^{n} |v_i| \leq \sum_{i=1}^{n} \sum_{j=1}^{n} p_{ij}|v_j|.$$

We may rearrange the finite sums:

$$|\lambda| \sum_{i=1}^{n} |v_i| \leq \sum_{j=1}^{n} \left(\sum_{i=1}^{n} p_{ij} \right) |v_j|.$$

But P is column-stochastic, so every column sums to one:

$$\sum_{i=1}^{n} p_{ij} = 1.$$

Therefore,

$$|\lambda| \sum_{i=1}^{n} |v_i| \leq \sum_{j=1}^{n} |v_j|.$$

Since $v \neq 0$, at least one component of v is nonzero.

Consequently,

$$\sum_{j=1}^{n} |v_j| > 0.$$

We may therefore divide both sides by this positive quantity without changing the inequality direction.

This gives

$$\boxed{|\lambda| \leq 1.}$$

Remember that λ was an arbitrary eigenvalue of P.

Thus, every eigenvalue lies inside or on the unit circle in the complex plane.

Since the spectral radius is the largest modulus among all eigenvalues,

$$\boxed{\rho(P) \leq 1.}$$

Conclusion. From Step 2, we established

$$\rho(P) \geq 1.$$

From Step 3, we established

$$\rho(P) \leq 1.$$

Combining these two inequalities,

$$1 \leq \rho(P) \leq 1.$$

Therefore,

$$\boxed{\rho(P) = 1.}$$

This completes the proof.

□

Remark 3.5. The vector $\mathbf{1}$ is always a right eigenvector of P^T corresponding to the eigenvalue 1.

However, it need not be a right eigenvector of P.

For a column-stochastic matrix, we know that

$$\mathbf{1}^T P = \mathbf{1}^T,$$

but this does not generally imply

$$P\mathbf{1} = \mathbf{1}.$$

We must therefore distinguish between left and right eigenvectors.

3.5 Stationary Probability Distributions

We now arrive at a central concept in the theory of Markov chains.

Suppose that a probability distribution x has the property that applying the transition matrix does not change it:

$$Px = x.$$

Such a distribution is called stationary.

Definition 3.6 (Stationary Distribution)**.** Let P be a column-stochastic matrix.

A vector $\pi \in \mathbb{R}^n$ is called a *stationary probability distribution* if

$$P\pi = \pi, \tag{45}$$

with

$$\pi \geq 0, \qquad \mathbf{1}^T\pi = 1.$$

Thus, a stationary distribution is a nonnegative eigenvector corresponding to the eigenvalue 1, normalized so that its components sum to one.

If the system begins in a stationary distribution, then

$$x^{(0)} = \pi$$

implies

$$x^{(1)} = P\pi = \pi.$$

Similarly,

$$x^{(2)} = P^2\pi = \pi,$$

and, more generally,

$$\boxed{P^k\pi = \pi \qquad \text{for every } k \geq 0.}$$

The distribution remains unchanged under repeated applications of P.

3.6 A Simple Example

Consider again the column-stochastic matrix

$$P = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}.$$

We want to find a stationary probability distribution

$$\pi = \begin{pmatrix} \pi_1 \\ \pi_2 \end{pmatrix}.$$

By definition,

$$P\pi = \pi.$$

Thus,

$$\begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix} \begin{pmatrix} \pi_1 \\ \pi_2 \end{pmatrix} = \begin{pmatrix} \pi_1 \\ \pi_2 \end{pmatrix}.$$

Expanding gives

$$0.8\pi_1 + 0.3\pi_2 = \pi_1,$$

and

$$0.2\pi_1 + 0.7\pi_2 = \pi_2.$$

Both equations reduce to

$$0.2\pi_1 = 0.3\pi_2.$$

Hence,

$$2\pi_1 = 3\pi_2.$$

We also require

$$\pi_1 + \pi_2 = 1.$$

Solving these equations gives

$$\boxed{\pi = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}.}$$

We may verify this directly:

$$\begin{aligned} P\pi &= \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix} \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix} \\ &= \begin{pmatrix} 0.60 \\ 0.40 \end{pmatrix} = \pi. \end{aligned}$$

Therefore, the stationary distribution assigns probability 0.6 to state 1 and probability 0.4 to state 2.

3.7 Existence of Stationary Distributions

We have shown that every column-stochastic matrix possesses the eigenvalue 1.

However, the existence of an eigenvalue alone does not immediately establish the existence of a corresponding nonnegative eigenvector.

We now show that every finite column-stochastic matrix has at least one stationary probability distribution.

Theorem 3.7 (Existence of a Stationary Distribution)**.** *Every column-stochastic matrix $P \in \mathbb{R}^{n\times n}$ has at least one stationary probability distribution.*

Proof. Let $x^{(0)}$ be any initial probability distribution.

For each integer $k \geq 1$, define the average

$$y^{(k)} = \frac{1}{k}\sum_{j=0}^{k-1} P^j x^{(0)}.$$

Each vector $P^j x^{(0)}$ is a probability distribution, because P maps the standard simplex into itself.

Therefore, their average $y^{(k)}$ also belongs to the standard simplex S.

Since S is compact, there exists a subsequence $(y^{(k_\ell)})$ converging to some vector $\pi \in S$.

Now observe that

$$\begin{aligned} Py^{(k)} - y^{(k)} &= \frac{1}{k}\left(\sum_{j=1}^{k} P^j x^{(0)} - \sum_{j=0}^{k-1} P^j x^{(0)}\right) \\ &= \frac{1}{k}\left(P^k x^{(0)} - x^{(0)}\right). \end{aligned}$$

Since both $P^k x^{(0)}$ and $x^{(0)}$ are probability distributions, their components lie in $[0, 1]$.

Consequently,

$$Py^{(k)} - y^{(k)} \longrightarrow 0.$$

In particular, this holds along the convergent subsequence.

Taking limits and using continuity of matrix multiplication gives

$$P\pi - \pi = 0.$$

Hence,

$$\boxed{P\pi = \pi.}$$

Since $\pi \in S$, it is a probability distribution.

Therefore, π is stationary.

□

Notice that the argument does not require P to have strictly positive entries.

It establishes existence even when some entries of P are zero.

3.8 Why Positivity Matters: Uniqueness and Convergence

The preceding theorem guarantees that every column-stochastic matrix has at least one stationary distribution.

However, it does not guarantee that the stationary distribution is unique.

Nor does it guarantee that the sequence

$$x^{(k)} = P^k x^{(0)}$$

converges as $k \to \infty$.

We examine two examples.

Example 1: Nonuniqueness. Consider

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Every probability distribution satisfies

$$P\pi = \pi.$$

Therefore, there are infinitely many stationary distributions.

In this case, the eigenvalue 1 has a two-dimensional eigenspace.

Example 2: Failure of convergence. Now consider

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This matrix is also column-stochastic.

Its eigenvalues are

$$\lambda_1 = 1, \qquad \lambda_2 = -1.$$

The stationary distribution is uniquely determined:

$$\pi = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}.$$

However, starting from

$$x^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

we obtain

$$x^{(1)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad x^{(2)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Thus, the probability distribution alternates between two states.

The sequence does not converge, even though the stationary distribution is unique.

The difficulty is that another eigenvalue, -1, has the same modulus as 1.

This is precisely the kind of difficulty that strict dominance eliminates.

3.9 The Perron–Frobenius Connection

We now bring together the results of this section and the Perron–Frobenius theorem established earlier.

Theorem 3.8 (Positive Stochastic Matrices)**.** *Let P be a strictly positive column-stochastic matrix.*

Then:

1. *The eigenvalue 1 is the spectral radius of P.*
2. *There exists a unique stationary probability distribution $\pi > 0$.*
3. *Every other eigenvalue $\lambda \neq 1$ satisfies*

$$|\lambda| < 1.$$

4. *For every initial probability distribution $x^{(0)}$,*

$$\lim_{k \to \infty} P^k x^{(0)} = \pi.$$

Proof. Because P is column-stochastic, we have already proved that

$$\rho(P) = 1.$$

Since $P > 0$, the Perron–Frobenius theorem guarantees a strictly positive eigenvector associated with this eigenvalue.

The eigenvalue 1 is algebraically simple, and every other eigenvalue has modulus strictly less than one.

Normalizing the positive eigenvector so that its components sum to one gives the unique stationary probability distribution π.

Finally, because 1 is algebraically simple and all other eigenvalues lie strictly inside the unit circle, the remaining terms in the Jordan decomposition of P^k decay to zero.

Thus, $P^k x^{(0)}$ converges to a scalar multiple of π.

Since every iterate is a probability distribution and π is normalized, that scalar must equal one.

Therefore,

$$\boxed{\lim_{k\to\infty} P^k x^{(0)} = \pi.}$$

□

This theorem provides the mathematical foundation for an important computational principle:

Repeated multiplication by a positive stochastic matrix eventually produces its unique stationary probability distribution, regardless of the initial probability distribution.

3.10 Connection to Web-Page Ranking

We conclude this section by returning to the eigenvalue problem

$$P\pi = \pi.$$

Suppose the states of our system represent web pages rather than physical locations or abstract states.

Imagine a person browsing the internet by repeatedly following hyperlinks.

We can interpret p_{ij} as the probability that the person moves from page j to page i.

If the browsing process has a unique stationary distribution

$$\pi = \begin{pmatrix} \pi_1 \\ \vdots \\ \pi_n \end{pmatrix},$$

then π_i represents the long-run probability of finding the browser at page i.

This suggests a method of ranking web pages.

A page with a larger stationary probability could be assigned a higher importance score.

Thus, the problem of ranking web pages can be transformed into the eigenvalue problem

$$\boxed{P\pi = \pi.}$$

However, a real network of hyperlinks generally contains many missing connections, so its transition matrix contains numerous zero entries.

Such a matrix need not satisfy the strict positivity assumption required by the version of Perron–Frobenius proved in this note.

Furthermore, some pages have no outgoing links, and certain groups of pages may form isolated or cyclic structures.

These difficulties motivate the construction of Google's PageRank algorithm and its subsequent modifications.

We investigate these ideas in the next section.

4 From Hyperlinks to Google's PageRank Algorithm

In the preceding section, we established that the stationary probability distribution of a Markov chain is an eigenvector corresponding to the eigenvalue one.

We also observed that, under suitable conditions, this stationary distribution is unique and can be obtained through repeated matrix multiplication.

We now investigate how these mathematical results can be used to assign importance scores to web pages.

The central idea behind PageRank is that hyperlinks between web pages contain information about their relative importance.

A hyperlink from one page to another may be interpreted as a recommendation or vote.

However, not all hyperlinks should necessarily contribute the same amount of importance.

We will develop this idea gradually, beginning with a simple directed graph and eventually arriving at an eigenvalue problem.

4.1 Representing the Web as a Directed Graph

Consider a collection of n web pages, which we label

$$1, 2, \ldots, n.$$

A web page may contain hyperlinks pointing to other pages in the collection.

We can represent this structure by a directed graph.

Each web page corresponds to a vertex, and a hyperlink from page j to page i corresponds to a directed edge from vertex j to vertex i.

The direction is important.

A hyperlink from page j to page i does not imply that page i contains a hyperlink back to page j.

We may encode the entire link structure in a matrix.

Definition 4.1 (Adjacency Matrix)**.** Suppose a collection of n web pages is represented by a directed graph.

The adjacency matrix $A = (a_{ij}) \in \mathbb{R}^{n \times n}$ is defined by

$$a_{ij} = \begin{cases} 1, & \text{if page } j \text{ links to page } i, \\ 0, & \text{otherwise.} \end{cases} \tag{46}$$

Notice our convention:

> The column index j identifies the page from which a hyperlink originates, while the row index i identifies the page to which the hyperlink points.

We choose this convention because it agrees with the column-stochastic matrices introduced in the preceding section.

4.2 An Example with Three Web Pages

Suppose our web consists of three pages with the following hyperlink structure:

1. Page 1 links to pages 2 and 3.
2. Page 2 links to page 3.
3. Page 3 links to page 1.

The corresponding directed edges are

$$1 \longrightarrow 2, \qquad 1 \longrightarrow 3,$$

$$2 \longrightarrow 3, \qquad 3 \longrightarrow 1.$$

Let us construct the adjacency matrix carefully.

Since page 1 links to pages 2 and 3, the first column is

$$\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Since page 2 links only to page 3, the second column is

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Finally, since page 3 links only to page 1, the third column is

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore,

$$\boxed{A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}.} \tag{47}$$

We can now use ordinary matrix multiplication to extract information about the network.

4.3 The Simplest Ranking Method: Counting Incoming Links

A natural first attempt at ranking web pages is to count the number of hyperlinks pointing to each page.

A page receiving more incoming links might be considered more important.

For our adjacency matrix, the number of incoming links to page i is simply the sum of the entries in row i.

We can calculate all these row sums simultaneously by multiplying A by the all-ones vector.

Let

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Then

$$A\mathbf{1} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

Thus, page 1 receives one incoming hyperlink, page 2 receives one, and page 3 receives two.

According to this ranking method, page 3 is the most important.

However, this method has a serious limitation.

Every hyperlink contributes exactly one vote, regardless of the page from which it originates.

Should a hyperlink from a page containing hundreds of outgoing links have the same importance as a hyperlink from a page containing only one?

It is reasonable to expect that these two links should contribute different amounts.

This motivates our first improvement.

4.4 The Weighted Linking Matrix

Suppose page j contains m_j outgoing hyperlinks.

Instead of assigning one full vote to every outgoing link, we allow page j to distribute one unit of voting strength equally among all its outgoing links.

Thus, each outgoing hyperlink from page j receives weight

$$\frac{1}{m_j}.$$

For the moment, we assume that every page has at least one outgoing hyperlink, so $m_j > 0$.

We define the following matrix.

Definition 4.2 (Weighted Linking Matrix)**.** Let $A = (a_{ij})$ be the adjacency matrix of a directed web graph, and suppose every page has at least one outgoing link.

For each page j, define

$$m_j = \sum_{i=1}^{n} a_{ij}.$$

The weighted linking matrix $W = (w_{ij})$ is defined by

$$w_{ij} = \frac{a_{ij}}{m_j}. \tag{48}$$

Thus,

$$w_{ij} = \begin{cases} \frac{1}{m_j}, & \text{if page } j \text{ links to page } i, \\ 0, & \text{otherwise.} \end{cases}$$

The entries in column j describe how page j distributes its voting strength.

Let us return to our three-page example.

Page 1 has two outgoing links, while pages 2 and 3 each have one.

Therefore,

$$m_1 = 2, \qquad m_2 = 1, \qquad m_3 = 1.$$

Dividing each column of the adjacency matrix by its corresponding number of outgoing links, we obtain

$$\boxed{W = \begin{pmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 1 & 0 \end{pmatrix}.} \tag{49}$$

Observe that every column of W sums to one:

$$\frac{1}{2} + \frac{1}{2} = 1, \qquad 1 = 1, \qquad 1 = 1.$$

More generally, for any column j,

$$\begin{aligned} \sum_{i=1}^{n} w_{ij} &= \sum_{i=1}^{n} \frac{a_{ij}}{m_j} \\ &= \frac{1}{m_j} \sum_{i=1}^{n} a_{ij} \\ &= \frac{m_j}{m_j} \\ &= 1. \end{aligned}$$

Since $w_{ij} \geq 0$, we conclude that

$$\boxed{W \text{ is column-stochastic.}}$$

This is our first important connection between hyperlinks and stochastic matrices.

4.5 Why Weighted Votes Are Still Not Enough

We have improved our original ranking scheme by allowing each page to distribute one unit of voting strength among its outgoing links.

However, another problem remains.

Suppose two pages each contain exactly one outgoing hyperlink.

One hyperlink comes from a highly influential page, while the other comes from a relatively unknown page.

Under our weighted linking scheme, both hyperlinks contribute exactly one unit of weight.

But we might reasonably expect a recommendation from an important page to carry greater influence.

This suggests a second improvement:

> A page should receive greater importance when it is linked to by other important pages.

At first, this idea appears circular.

How can we determine which pages are important if their importance depends on the importance of other pages?

The key observation is that this apparent circularity can be expressed as a system of linear equations.

4.6 Deriving the PageRank Eigenvalue Equation

Let r_i denote the importance score assigned to page i.

Collect all these scores into a vector

$$r = \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{pmatrix}.$$

We now adopt the following ranking principle.

The importance of page i should equal the sum of the importance contributed by all pages linking to it.

If page j has importance r_j and distributes this importance equally among its m_j outgoing links, then its contribution to page i is

$$w_{ij} r_j.$$

Therefore, the importance score of page i should satisfy

$$r_i = \sum_{j=1}^{n} w_{ij} r_j, \qquad i = 1, \ldots, n. \tag{50}$$

Notice that this is a system of n linear equations.

Writing all equations simultaneously gives

$$\begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{pmatrix} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ w_{21} & \cdots & w_{2n} \\ \vdots & \ddots & \vdots \\ w_{n1} & \cdots & w_{nn} \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{pmatrix}.$$

In matrix notation,

$$\boxed{Wr = r.} \tag{51}$$

This is precisely an eigenvalue equation!

The desired ranking vector is a right eigenvector of W corresponding to the eigenvalue 1.

Thus, a seemingly circular definition of web-page importance has become a familiar problem from linear algebra.

Since W is column-stochastic, we already know that $\rho(W) = 1$ and that at least one stationary probability distribution exists.

We can therefore normalize the ranking vector by requiring

$$r_i \geq 0, \qquad \sum_{i=1}^{n} r_i = 1.$$

This gives the scores a probabilistic interpretation.

4.7 Computing the Rankings: A Three-Page Example

Let us return to the weighted linking matrix

$$W = \begin{pmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 1 & 0 \end{pmatrix}.$$

We seek a nonnegative vector

$$r = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}$$

satisfying

$$Wr = r, \qquad r_1 + r_2 + r_3 = 1.$$

Expanding the matrix equation gives

$$\begin{pmatrix} r_3 \\ \frac{1}{2}r_1 \\ \frac{1}{2}r_1 + r_2 \end{pmatrix} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}.$$

Hence,

$$\begin{aligned} r_1 &= r_3, \\ r_2 &= \frac{1}{2} r_1, \\ r_3 &= \frac{1}{2} r_1 + r_2. \end{aligned}$$

The third equation follows from the first two, so we use the normalization condition to determine the scores.

Substituting

$$r_2 = \frac{1}{2} r_1, \qquad r_3 = r_1$$

into

$$r_1 + r_2 + r_3 = 1$$

gives

$$r_1 + \frac{1}{2} r_1 + r_1 = 1.$$

Therefore,

$$\frac{5}{2} r_1 = 1,$$

so

$$r_1 = \frac{2}{5}.$$

Consequently,

$$r_2 = \frac{1}{5}, \qquad r_3 = \frac{2}{5}.$$

Our ranking vector is

$$\boxed{r = \begin{pmatrix} 0.4 \\ 0.2 \\ 0.4 \end{pmatrix}.} \tag{52}$$

Thus, pages 1 and 3 receive equal importance scores of 0.4, while page 2 receives a score of 0.2.

Notice that pages 1 and 2 each receive one incoming hyperlink, yet they have different importance scores.

The reason is that page 1 receives its link from page 3, which passes all its importance to page 1.

Page 2, by contrast, receives its link from page 1, which divides its importance between two destinations.

This illustrates why PageRank captures more information than simply counting incoming links.

4.8 The Random-Surfer Interpretation

There is another useful way to understand the ranking vector.

Imagine a person browsing the internet by repeatedly following hyperlinks.

Whenever the person arrives at a page, they choose one of its outgoing hyperlinks uniformly at random.

For example, if the person arrives at page 1 in our three-page network, they have probability $\frac{1}{2}$ of moving to page 2 and probability $\frac{1}{2}$ of moving to page 3.

If the person arrives at page 2, they move to page 3 with probability one.

Thus, the weighted linking matrix W is also the transition matrix of a Markov chain.

If the probability distribution after k transitions is $x^{(k)}$, then

$$x^{(k+1)} = Wx^{(k)}.$$

A stationary distribution r satisfies

$$Wr = r.$$

When the Markov chain converges to a unique stationary distribution, the component r_i represents the long-run probability of finding the surfer at page i.

This gives a natural interpretation of the ranking scores:

> A web page is considered important if a random surfer is likely to visit it in the long run.

The matrix formulation and the random-surfer interpretation therefore describe the same mathematical model.

4.9 Limitations of the Basic Model

The basic PageRank model has led us to the eigenvalue problem

$$Wr = r.$$

However, several difficulties remain when we attempt to apply this method to a realistic web graph.

First, some web pages have no outgoing hyperlinks.

Such pages are commonly called *dangling pages*.

If page j has no outgoing links, then

$$m_j = 0.$$

Consequently, the formula

$$w_{ij} = \frac{a_{ij}}{m_j}$$

is undefined.

We must therefore modify the construction of W before it can represent a valid transition matrix.

Second, even after dealing with dangling pages, the resulting matrix may have many zero entries.

The Perron–Frobenius theorem for strictly positive matrices therefore cannot be applied directly.

As we saw in the preceding section, a nonnegative stochastic matrix may have multiple stationary distributions.

Even when the stationary distribution is unique, repeated multiplication need not converge.

These difficulties raise two important questions:

1. How can we modify the hyperlink matrix so that it represents a valid stochastic process even when dangling pages are present?
2. How can we guarantee that the resulting ranking vector is unique and that the numerical iteration converges to it?

These questions motivate the modified PageRank algorithm, which introduces a teleportation mechanism controlled by a damping parameter.

We develop this modification and investigate its mathematical justification in the next section.

5 The Modified PageRank Algorithm: Dangling Nodes, Teleportation, and Convergence

In the preceding section, we developed the basic PageRank algorithm by representing hyperlinks as a directed graph and constructing a weighted linking matrix.

We showed that the desired ranking vector satisfies the eigenvalue equation

$$Wr = r.$$

This equation has a natural probabilistic interpretation: the ranking vector represents a stationary distribution of a random surfer moving between web pages by following hyperlinks.

However, the basic model presents several mathematical difficulties.

Some pages contain no outgoing links, so the weighted linking matrix may fail to be stochastic.

Furthermore, even when the matrix is stochastic, its stationary distribution may not be unique, and repeated matrix multiplication may fail to converge.

We now investigate how these difficulties can be resolved.

Our objective is to construct a modified transition matrix that satisfies the hypotheses of the Perron–Frobenius theorem for strictly positive matrices.

5.1 The Problem of Dangling Pages

Recall that the weighted linking matrix was defined by

$$w_{ij} = \frac{a_{ij}}{m_j},$$

where m_j denotes the number of outgoing hyperlinks from page j.

This formula requires

$$m_j > 0.$$

But not every web page contains outgoing hyperlinks.

A page having no outgoing links is called a *dangling page* or *dangling node*.

Suppose, for example, that we have three web pages with the following connections:

$$1 \longrightarrow 2, \qquad 2 \longrightarrow 1.$$

Page 3 has no outgoing links.

The adjacency matrix is

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The numbers of outgoing links are

$$m_1 = 1, \qquad m_2 = 1, \qquad m_3 = 0.$$

We can normalize the first two columns, but the third column presents a problem because $m_3 = 0$.

If we simply leave the third column equal to zero, we obtain

$$W = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Notice that the third column sums to zero, rather than one.

Consequently, W is not column-stochastic.

To understand why this matters, consider a probability distribution concentrated entirely on page 3:

$$x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Multiplication gives

$$Wx = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

The resulting vector is not a probability distribution, because its components sum to zero rather than one.

In probabilistic terms, the surfer has arrived at a page with no outgoing links and has no prescribed way to continue browsing.

We must therefore modify the transition rule.

5.2 Repairing the Dangling Columns

A simple solution is to allow the surfer to choose a new page uniformly at random whenever a dangling page is encountered.

If there are n pages, the probability of selecting any particular page is

$$\frac{1}{n}.$$

Thus, a dangling column is replaced by

$$\frac{1}{n}\mathbf{1},$$

where

$$\mathbf{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}.$$

We can express this construction mathematically.

For each page j, define

$$m_j = \sum_{i=1}^{n} a_{ij}.$$

We construct a new matrix $S = (s_{ij})$ by setting

$$s_{ij} = \begin{cases} \dfrac{a_{ij}}{m_j}, & m_j > 0, \\ \dfrac{1}{n}, & m_j = 0. \end{cases} \tag{53}$$

For our three-page example, we obtain

$$\boxed{S = \begin{pmatrix} 0 & 1 & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} \\ 0 & 0 & \frac{1}{3} \end{pmatrix}.} \tag{54}$$

Every entry is nonnegative, and every column sums to one.

Therefore, S is column-stochastic.

More generally, we obtain the following proposition.

Proposition 5.1. *The matrix S defined by* (53) *is column-stochastic.*

Proof. Every entry of S is nonnegative.

Now consider an arbitrary column j.

If $m_j > 0$, then

$$\sum_{i=1}^{n} s_{ij} = \frac{1}{m_j} \sum_{i=1}^{n} a_{ij} = 1.$$

If $m_j = 0$, then

$$\sum_{i=1}^{n} s_{ij} = \sum_{i=1}^{n} \frac{1}{n} = n \cdot \frac{1}{n} = 1.$$

Thus, every column sums to one.

Hence S is column-stochastic.

□

We have now obtained a valid transition matrix.

However, notice that S still contains zero entries.

Therefore, it need not be strictly positive.

Although we have repaired the dangling pages, we have not yet guaranteed a unique stationary distribution or convergence.

This motivates a second, more important modification.

5.3 The Random-Surfer Model with Teleportation

Consider again a person browsing the internet.

In our original model, the surfer always followed one of the hyperlinks on the current page.

This assumption is rather restrictive.

In practice, a person may stop following links and visit a completely different page.

For example, the person may type a new address into the browser or start a new search.

We can incorporate this behavior into our mathematical model.

Suppose that, at every step, the surfer makes one of two choices.

1. With probability d, the surfer follows a hyperlink according to the transition matrix S.
2. With probability $1 - d$, the surfer jumps to a page selected uniformly at random from the entire collection.

Here,

$$0 < d < 1.$$

The parameter d is called the *damping factor*.

For example, if

$$d = 0.85,$$

then the surfer follows the link-based transition rule with probability 0.85 and teleports with probability 0.15.

The value 0.85 is a commonly used illustrative choice.

The essential mathematical requirement, however, is simply

$$0 < d < 1.$$

We now translate this probabilistic description into a matrix equation.

5.4 Derivation of the Google Matrix

Recall that S is column-stochastic.

Thus, s_{ij} represents the probability of moving from page j to page i under the link-based transition rule.

With probability d, the surfer follows this rule.

Therefore, this behavior contributes probability

$$d\, s_{ij}$$

to the transition from page j to page i.

With probability $1 - d$, the surfer teleports.

Since the destination is chosen uniformly among n pages, the probability of teleporting specifically to page i is

$$\frac{1-d}{n}.$$

The total transition probability is therefore

$$g_{ij} = d\, s_{ij} + \frac{1-d}{n}. \tag{55}$$

We collect these probabilities into a matrix $G = (g_{ij})$.

Let

$$E = \frac{1}{n}\mathbf{1}\mathbf{1}^T.$$

Since

$$\mathbf{1}\mathbf{1}^T = \begin{pmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix},$$

the matrix E has every entry equal to $1/n$.

Therefore,

$$G = dS + (1-d)E.$$

Equivalently,

$$\boxed{G = dS + \frac{1-d}{n}\mathbf{1}\mathbf{1}^T.} \tag{56}$$

This matrix is called the *Google matrix*.

Notice that it is a convex combination of two transition matrices:

$$G = dS + (1-d)E.$$

The matrix S describes movement according to hyperlinks, whereas E describes uniform random teleportation.

The damping factor d determines the relative importance of these two mechanisms.

5.5 Why Is the Google Matrix Strictly Positive and Stochastic?

We now establish the mathematical properties of the Google matrix.

Theorem 5.2. *Let S be a column-stochastic matrix, and let $0 < d < 1$.*

Define

$$G = dS + \frac{1-d}{n}\mathbf{1}\mathbf{1}^T.$$

Then G is strictly positive and column-stochastic.

Proof. We prove the two properties separately.

Step 1: Strict positivity. From the definition of G, its entries are

$$g_{ij} = d\, s_{ij} + \frac{1-d}{n}.$$

Since S is column-stochastic,

$$s_{ij} \geq 0.$$

Furthermore,

$$0 < d < 1$$

implies

$$1 - d > 0.$$

Therefore,

$$g_{ij} \geq \frac{1-d}{n} > 0.$$

Hence every entry of G is strictly positive:

$$\boxed{G > 0.}$$

Notice that this conclusion holds even when S contains many zero entries.

The teleportation term ensures strict positivity.

Step 2: Column-stochasticity. We must show that every column of G sums to one.

Fix any column j.

Then

$$\begin{aligned}\sum_{i=1}^{n} g_{ij} &= \sum_{i=1}^{n} \left(d\, s_{ij} + \frac{1-d}{n} \right) \\ &= d \sum_{i=1}^{n} s_{ij} + \sum_{i=1}^{n} \frac{1-d}{n}.\end{aligned}$$

Because S is column-stochastic,

$$\sum_{i=1}^{n} s_{ij} = 1.$$

Also,

$$\sum_{i=1}^{n} \frac{1-d}{n} = 1 - d.$$

Therefore,

$$\begin{aligned}\sum_{i=1}^{n} g_{ij} &= d + (1-d) \\ &= 1.\end{aligned}$$

Thus, every column of G sums to one.

Since we already established $G > 0$, we conclude that G is strictly positive and column-stochastic.

□

This theorem provides the essential connection to the Perron–Frobenius theorem developed earlier in our note.

5.6 Existence and Uniqueness of the PageRank Vector

We can now apply the Perron–Frobenius theorem directly.

Theorem 5.3 (Existence and Uniqueness of PageRank)**.** *Let G be the Google matrix defined by* (56)*, where S is column-stochastic and $0 < d < 1$.*

Then there exists a unique probability distribution $r > 0$ satisfying

$$\boxed{Gr = r.} \tag{57}$$

Moreover, the eigenvalue 1 *is algebraically simple, and every other eigenvalue λ satisfies*

$$|\lambda| < 1.$$

Proof. We established that G is strictly positive and column-stochastic.

Since G is column-stochastic, our earlier result gives

$$\rho(G) = 1.$$

Since $G > 0$, the Perron–Frobenius theorem guarantees that the spectral radius is an algebraically simple eigenvalue with a strictly positive eigenvector.

Therefore, there exists a vector $r > 0$ such that

$$Gr = r.$$

Because $r > 0$, the sum of its components is strictly positive.

We may therefore normalize r so that

$$\sum_{i=1}^{n} r_i = 1.$$

This gives a stationary probability distribution.

Furthermore, the Perron–Frobenius theorem establishes that the eigenspace corresponding to the eigenvalue one is one-dimensional.

Thus, any other eigenvector associated with the eigenvalue one must be a scalar multiple of r.

Since we require the components to sum to one, the normalization determines that scalar uniquely.

Hence the stationary probability distribution is unique.

Finally, strict dominance of the Perron eigenvalue gives

$$|\lambda| < 1$$

for every eigenvalue $\lambda \neq 1$.

□

The conclusion is important.

The basic hyperlink matrix may have multiple stationary distributions, but the modified Google matrix possesses exactly one stationary probability distribution.

This uniquely determined vector is the PageRank vector of our model.

5.7 A Numerical Example with a Dangling Page

We now return to our three-page network.

Recall that pages 1 and 2 link to each other, while page 3 is dangling.

After repairing the dangling column, we obtained

$$S = \begin{pmatrix} 0 & 1 & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} \\ 0 & 0 & \frac{1}{3} \end{pmatrix}.$$

Let us choose

$$d = \frac{4}{5} = 0.8.$$

Then

$$1 - d = \frac{1}{5}.$$

Since there are three pages,

$$G = \frac{4}{5}S + \frac{1}{15}\mathbf{1}\mathbf{1}^T.$$

Substituting the entries of S gives

$$G = \frac{4}{5}\begin{pmatrix} 0 & 1 & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} \\ 0 & 0 & \frac{1}{3} \end{pmatrix} + \frac{1}{15}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Therefore,

$$\boxed{G = \begin{pmatrix} \frac{1}{15} & \frac{13}{15} & \frac{1}{3} \\ \frac{13}{15} & \frac{1}{15} & \frac{1}{3} \\ \frac{1}{15} & \frac{1}{15} & \frac{1}{3} \end{pmatrix}.} \tag{58}$$

Notice that all entries are strictly positive, and every column sums to one.

We now seek a probability distribution

$$r = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}$$

satisfying

$$Gr = r.$$

Expanding gives

$$\begin{aligned} r_1 &= \frac{1}{15}r_1 + \frac{13}{15}r_2 + \frac{1}{3}r_3, \\ r_2 &= \frac{13}{15}r_1 + \frac{1}{15}r_2 + \frac{1}{3}r_3, \\ r_3 &= \frac{1}{15}r_1 + \frac{1}{15}r_2 + \frac{1}{3}r_3. \end{aligned}$$

By symmetry between pages 1 and 2, we expect

$$r_1 = r_2.$$

To verify this, subtract the second equation from the first:

$$r_1 - r_2 = -\frac{4}{5}(r_1 - r_2).$$

Therefore,

$$\frac{9}{5}(r_1 - r_2) = 0,$$

so

$$r_1 = r_2.$$

Now consider the third equation:

$$r_3 = \frac{1}{15}(r_1 + r_2) + \frac{1}{3}r_3.$$

Rearranging,

$$\frac{2}{3}r_3 = \frac{1}{15}(r_1 + r_2).$$

Thus,

$$r_1 + r_2 = 10r_3.$$

Since r is a probability distribution,

$$r_1 + r_2 + r_3 = 1.$$

Consequently,

$$10r_3 + r_3 = 1,$$

which gives

$$r_3 = \frac{1}{11}.$$

Therefore,

$$r_1 + r_2 = \frac{10}{11}.$$

Since $r_1 = r_2$,

$$r_1 = r_2 = \frac{5}{11}.$$

Our final PageRank vector is

$$\boxed{r = \begin{pmatrix} \frac{5}{11} \\ \frac{5}{11} \\ \frac{1}{11} \end{pmatrix}.} \tag{59}$$

Thus, pages 1 and 2 receive approximately 45.45% of the stationary probability each, while page 3 receives approximately 9.09%.

Page 3 receives a positive ranking score even though it has no incoming hyperlinks.

This occurs because the teleportation mechanism allows the surfer to reach every page.

The example illustrates how the modified PageRank algorithm produces a unique positive ranking vector, even for a network containing a dangling page.

5.8 Computing PageRank by Power Iteration

We have established the existence and uniqueness of the PageRank vector.

However, for a web containing millions of pages, computing the characteristic polynomial of G would be impractical.

Instead, we use an iterative method.

Choose an initial probability distribution

$$x^{(0)} \geq 0, \qquad \mathbf{1}^T x^{(0)} = 1.$$

Then define

$$x^{(k+1)} = Gx^{(k)}, \qquad k = 0, 1, 2, \ldots \tag{60}$$

Because G is column-stochastic, each iterate remains a probability distribution.

Since G is strictly positive, the Perron–Frobenius theorem guarantees that the eigenvalue one is algebraically simple and strictly dominant.

Consequently,

$$\boxed{x^{(k)} \longrightarrow r \qquad \text{as } k \to \infty,} \tag{61}$$

where r is the unique stationary probability distribution.

Thus, repeated matrix multiplication provides a practical method for approximating the PageRank vector.

In the next subsection, we establish an even stronger convergence property that follows from the special structure of the Google matrix.

5.9 Why Teleportation Improves Convergence

The Perron–Frobenius theorem guarantees convergence, but the structure of the Google matrix allows us to obtain a quantitative estimate.

Recall that

$$G = dS + \frac{1-d}{n}\mathbf{1}\mathbf{1}^T.$$

Let r be the stationary PageRank vector:

$$Gr = r.$$

Suppose $x^{(0)}$ is any initial probability distribution, and let

$$x^{(k+1)} = Gx^{(k)}.$$

Define the error vector

$$e^{(k)} = x^{(k)} - r.$$

Since both $x^{(k)}$ and r are probability distributions,

$$\mathbf{1}^T x^{(k)} = 1, \qquad \mathbf{1}^T r = 1.$$

Therefore,

$$\mathbf{1}^T e^{(k)} = 0. \tag{62}$$

Now observe that

$$\begin{aligned} e^{(k+1)} &= x^{(k+1)} - r \\ &= Gx^{(k)} - Gr \\ &= G(x^{(k)} - r) \\ &= Ge^{(k)}. \end{aligned}$$

Substituting the definition of G,

$$e^{(k+1)} = dSe^{(k)} + \frac{1-d}{n}\mathbf{1}\mathbf{1}^T e^{(k)}.$$

But (62) gives

$$\mathbf{1}^T e^{(k)} = 0.$$

Hence the teleportation term vanishes:

$$\boxed{e^{(k+1)} = dSe^{(k)}.} \tag{63}$$

We now estimate the size of the error.

Recall that the vector 1-norm is

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|.$$

Because S is column-stochastic,

$$\|Sy\|_1 \leq \|y\|_1$$

for every vector y.

Indeed,

$$\begin{aligned} \|Sy\|_1 &= \sum_{i=1}^{n} \left| \sum_{j=1}^{n} s_{ij} y_j \right| \\ &\leq \sum_{i=1}^{n} \sum_{j=1}^{n} s_{ij} |y_j| \\ &= \sum_{j=1}^{n} \left(\sum_{i=1}^{n} s_{ij} \right) |y_j| \\ &= \sum_{j=1}^{n} |y_j| \\ &= \|y\|_1. \end{aligned}$$

Applying this inequality to the error iteration,

$$\begin{aligned} \|e^{(k+1)}\|_1 &= d\|Se^{(k)}\|_1 \\ &\leq d\|e^{(k)}\|_1. \end{aligned}$$

Repeating the argument,

$$\|e^{(k)}\|_1 \leq d^k \|e^{(0)}\|_1.$$

Since $0 < d < 1$,

$$d^k \longrightarrow 0.$$

Therefore,

$$\boxed{\|x^{(k)} - r\|_1 \leq d^k \|x^{(0)} - r\|_1.} \tag{64}$$

This establishes convergence directly, without requiring the computation of any eigenvalues.

It also provides a useful estimate of the convergence rate.

A smaller damping factor generally gives a stronger worst-case contraction bound.

However, reducing d also increases the influence of teleportation relative to the hyperlink structure.

Thus, the choice of d involves a tradeoff between the contribution of link information and computational convergence.

5.10 An Efficient Form of the Iteration

The Google matrix is

$$G = dS + \frac{1-d}{n}\mathbf{1}\mathbf{1}^T.$$

At first, this formula suggests that we must construct a dense matrix, because the teleportation term has no zero entries.

For a large web graph, this would require an enormous amount of memory.

Fortunately, this is unnecessary.

Recall the iteration

$$x^{(k+1)} = Gx^{(k)}.$$

Substituting the definition of G,

$$x^{(k+1)} = dSx^{(k)} + \frac{1-d}{n}\mathbf{1}\mathbf{1}^T x^{(k)}.$$

Since $x^{(k)}$ is a probability distribution,

$$\mathbf{1}^T x^{(k)} = 1.$$

Therefore,

$$\boxed{x^{(k+1)} = dSx^{(k)} + \frac{1-d}{n}\mathbf{1}.} \tag{65}$$

This formula is computationally important.

We do not need to store the full Google matrix G.

Instead, we compute the matrix-vector product $Sx^{(k)}$ and add the teleportation vector.

Since the hyperlink structure is typically sparse, the matrix S can be represented efficiently using sparse data structures.

Moreover, the dangling columns can also be handled implicitly, without explicitly storing their uniform entries.

We will explore these implementation details in the numerical section.

5.11 Personalized PageRank

So far, we have assumed that teleportation selects every page with equal probability.

This is called *uniform teleportation*.

However, the same mathematical framework allows us to assign different teleportation probabilities to different pages.

Let

$$v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$$

be a probability distribution:

$$v_i \geq 0, \qquad \sum_{i=1}^{n} v_i = 1.$$

The component v_i represents the probability that a teleporting surfer selects page i.

We replace the uniform teleportation matrix

$$\frac{1}{n}\mathbf{1}\mathbf{1}^T$$

by

$$v\mathbf{1}^T.$$

Thus, the generalized Google matrix becomes

$$\boxed{G_v = dS + (1-d)v\mathbf{1}^T.} \tag{66}$$

This construction is called *personalized PageRank*.

It allows the ranking to reflect a specified preference distribution rather than treating all pages equally during teleportation.

Notice that

$$(G_v)_{ij} = d\, s_{ij} + (1-d)v_i.$$

Since $S \geq 0$ and $v \geq 0$, every entry of G_v is nonnegative.

Furthermore,

$$\begin{aligned}\sum_{i=1}^{n}(G_v)_{ij} &= d\sum_{i=1}^{n} s_{ij} + (1-d)\sum_{i=1}^{n} v_i \\ &= d + (1-d) \\ &= 1.\end{aligned}$$

Hence G_v is column-stochastic.

If, in addition,

$$v > 0,$$

then

$$(G_v)_{ij} \geq (1-d)v_i > 0.$$

Therefore,

$$G_v > 0.$$

The Perron–Frobenius theorem then guarantees a unique strictly positive stationary probability distribution.

More generally, even if some components of v are zero, the contraction argument used in the preceding subsection still guarantees a unique stationary probability distribution and convergence of the iteration for $0 < d < 1$.

In that more general case, however, the stationary distribution need not have every component strictly positive.

The personalized PageRank iteration is

$$\boxed{x^{(k+1)} = dSx^{(k)} + (1-d)v.} \tag{67}$$

This formulation provides a flexible framework for ranking objects in directed networks.

5.12 Mathematical Significance of the Modification

We conclude this section by summarizing what the modified PageRank algorithm accomplishes.

The original hyperlink matrix represents the structure of a directed network.

However, dangling pages may prevent it from being stochastic.

Repairing dangling columns produces a valid transition matrix S.

Teleportation then gives the modified Google matrix

$$G = dS + \frac{1-d}{n}\mathbf{1}\mathbf{1}^T.$$

For $0 < d < 1$, this matrix is strictly positive and column-stochastic.

Consequently, the Perron–Frobenius theorem guarantees a unique positive stationary probability distribution.

The resulting PageRank vector satisfies

$$\boxed{Gr = r, \qquad r > 0, \qquad \mathbf{1}^T r = 1.}$$

Furthermore, power iteration converges from every initial probability distribution.

The convergence estimate

$$\|x^{(k)} - r\|_1 \leq d^k \|x^{(0)} - r\|_1$$

provides a direct quantitative explanation of this behavior.

An especially important observation is that teleportation introduces strictly positive matrix entries *by construction*.

This should not be confused with floating-point rounding errors.

In numerical computation, a zero matrix entry does not automatically become machine epsilon.

The positivity of the Google matrix arises from the explicit teleportation term, not from numerical rounding.

Thus, the PageRank modification illustrates how mathematical modeling can transform a difficult eigenvalue problem into one with guaranteed existence, uniqueness, and convergence properties.

6 Numerical Implementation and Computational Results

In this section, we investigate the numerical behavior of Google's PageRank algorithm using MATLAB and a real directed web graph.

Our objectives are to verify the mathematical properties established in the preceding sections, examine the convergence of power iteration, and investigate how the damping factor affects computational efficiency.

6.1 Dataset Description

We use the Stanford Web Graph dataset from the Stanford Network Analysis Project (SNAP).

The dataset represents a directed network of web pages collected from Stanford University's website in 2002.

The network contains

$$n = 281{,}903$$

nodes and

$$m = 2{,}312{,}497$$

directed edges.

Each node represents a web page, and each directed edge represents a hyperlink from one page to another.

The dataset was downloaded automatically by our MATLAB program, and the edge list was imported to construct the adjacency matrix.

Because the web graph is directed, we follow the column convention developed earlier:

$$A_{ij} = 1$$

whenever page j contains a hyperlink pointing to page i.

The resulting sparse adjacency matrix contains 2,312,497 nonzero entries.

Its density is

$$\begin{aligned} \text{Density} &= \frac{\text{nnz}(A)}{n^2} \times 100\% \\ &\approx 0.002910\%. \end{aligned}$$

Thus, more than 99.99% of the matrix entries are zero.

This observation motivates the use of MATLAB's sparse matrix representation.

6.2 Construction of the Transition Matrix

The number of outgoing links from page j is

$$m_j = \sum_{i=1}^{n} A_{ij}.$$

For every page with at least one outgoing link, we define

$$W_{ij} = \frac{A_{ij}}{m_j}.$$

Our dataset contains 172 dangling pages, corresponding to approximately 0.06% of all nodes.

These pages have no outgoing hyperlinks and therefore require special treatment.

As established previously, we replace the transition probabilities of each dangling page with a uniform distribution.

However, explicitly storing these additional entries would unnecessarily increase the memory requirements.

Our MATLAB implementation therefore handles the probability mass associated with dangling pages implicitly during power iteration.

6.3 Sparse PageRank Iteration

Recall that the modified Google matrix is

$$G = dS + (1-d)\frac{1}{n}\mathbf{1}\mathbf{1}^T.$$

The desired PageRank vector r satisfies

$$Gr = r, \qquad r > 0, \qquad \mathbf{1}^T r = 1.$$

Rather than constructing the dense matrix G, we use an equivalent sparse iteration.

Let W be the sparse weighted linking matrix whose dangling columns remain zero.

At iteration k, define the total probability mass at dangling pages by

$$c_k = \sum_{\substack{j \\ m_j=0}} x_j^{(k)}.$$

The update becomes

$$\boxed{x^{(k+1)} = dWx^{(k)} + \frac{dc_k + 1 - d}{n}\mathbf{1}.} \tag{68}$$

The term dc_k/n redistributes the probability mass associated with dangling pages.

The term $(1-d)/n$ represents uniform teleportation.

Together, these terms ensure that the iterates remain valid probability distributions.

We initialize the algorithm using the uniform distribution

$$x^{(0)} = \frac{1}{n}\mathbf{1}.$$

At each iteration, we calculate

$$\delta_k = \|x^{(k+1)} - x^{(k)}\|_1. \tag{69}$$

The algorithm terminates when

$$\delta_k < 10^{-10},$$

subject to a maximum of 700 iterations.

6.4 Effect of the Damping Factor

We tested three damping factors:

$$d \in \{0.70, 0.85, 0.95\}.$$

All three experiments converged within the prescribed tolerance.

Table 1 summarizes the numerical results.

Table 1: PageRank convergence on the Stanford Web Graph using MATLAB.

d	Iterations	Time (s)	L^1 residual
0.70	55	0.5547	5.69×10^{-11}
0.85	118	1.2989	8.39×10^{-11}
0.95	368	3.2516	9.46×10^{-11}

The reported residual is

$$\|Gr - r\|_1,$$

where r denotes the final numerical approximation.

We observe that increasing the damping factor leads to a substantial increase in the number of iterations.

For $d = 0.70$, the method requires only 55 iterations.

For $d = 0.85$, the iteration count increases to 118.

For $d = 0.95$, the algorithm requires 368 iterations.

Thus, increasing d from 0.70 to 0.95 increases the number of iterations by a factor of approximately 6.7.

This behavior agrees with our earlier convergence estimate

$$\|x^{(k)} - r\|_1 \leq d^k \|x^{(0)} - r\|_1.$$

As d approaches one, the factor d^k decreases more slowly.

Consequently, the theoretical worst-case convergence guarantee becomes weaker.

Figure 1 illustrates the convergence behavior for the three damping factors.

6.5 Numerical Verification

Our implementation also verifies that the computed PageRank vectors satisfy the expected probability constraints.

For all three damping factors,

$$\sum_{i=1}^{n} r_i \approx 1.$$

The smallest computed components were

$$\begin{aligned} d = 0.70: &\quad \min_i r_i = 1.065366 \times 10^{-6}, \\ d = 0.85: &\quad \min_i r_i = 5.333686 \times 10^{-7}, \\ d = 0.95: &\quad \min_i r_i = 1.784917 \times 10^{-7}. \end{aligned}$$

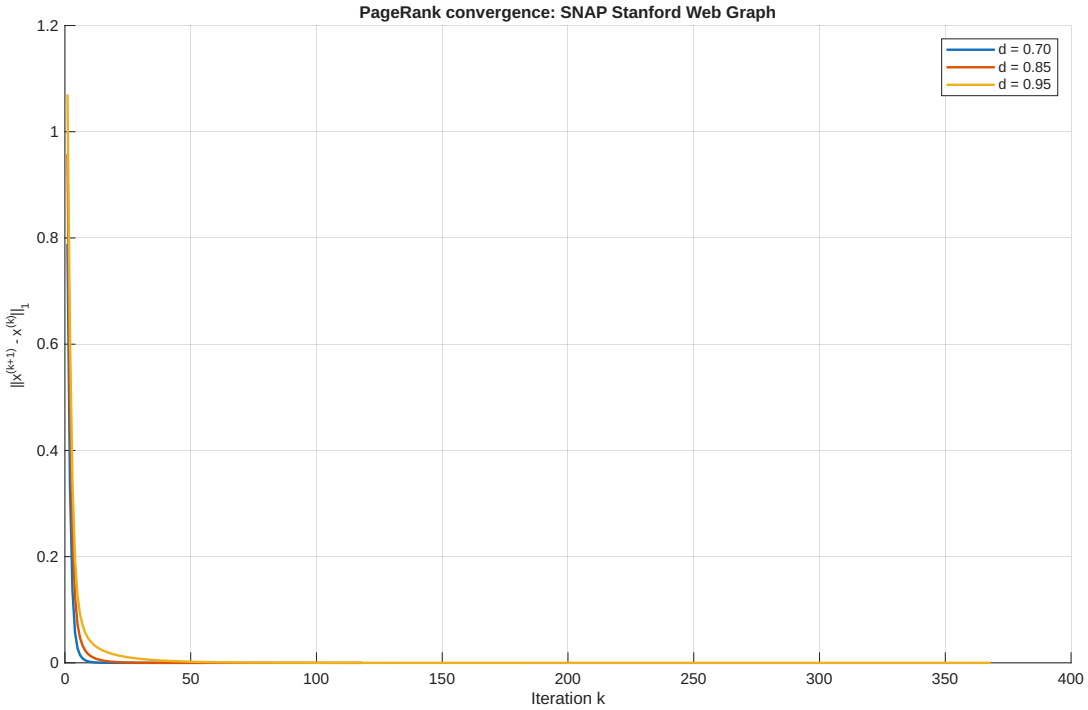


Figure 1: Convergence of PageRank power iteration for three damping factors. The plotted quantity is the L^1 difference between successive iterates.

Every computed component was therefore strictly positive.

The fixed-point residuals were also below 10^{-10}.

These observations are consistent with the mathematical properties established by the Perron–Frobenius theorem.

In particular, the Google matrix possesses a unique strictly positive stationary probability distribution, and the numerical iteration converges toward it.

6.6 Ranking the Web Pages

We now examine the PageRank scores computed using the damping factor

$$d = 0.85.$$

The highest-ranking node has SNAP identifier

$$89073$$

and PageRank score

$$r_{89073} \approx 0.011303.$$

The second and third highest-ranking nodes have scores approximately

$$0.009268 \quad \text{and} \quad 0.008297,$$

respectively.

The fourth highest score is approximately

$$0.003023.$$

Thus, the first three nodes receive substantially higher stationary probabilities than the remaining nodes in the top-twenty ranking.

Figure 2 displays the twenty largest PageRank scores.

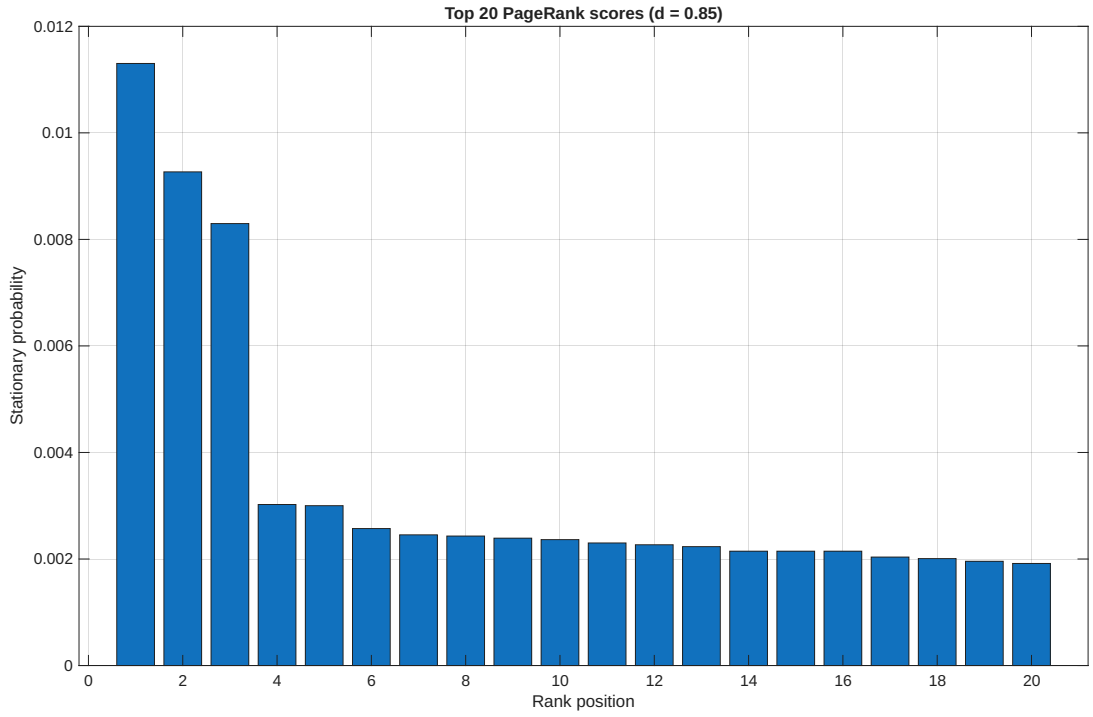


Figure 2: The twenty largest PageRank scores for the Stanford Web Graph with damping factor $d = 0.85$. The horizontal axis indicates ranking position, with position one corresponding to the highest-scoring node.

These scores can be interpreted as stationary probabilities in the modified random-surfer model.

For example, the highest-ranking node receives approximately 1.13% of the total stationary probability.

Although the graph contains more than 280,000 nodes, the stationary distribution is not uniform.

Instead, it reflects the hyperlink structure together with the teleportation mechanism.

6.7 Computational Discussion

The experiment demonstrates the practical importance of sparse matrix methods.

A dense matrix containing n^2 double-precision entries would require

$$8n^2 \text{ bytes}$$

of memory.

For $n = 281{,}903$, this corresponds to approximately

$$636 \text{ GB}$$

of storage in decimal units.

Such a representation would be unnecessarily expensive for this dataset.

Our implementation instead stores only the nonzero entries of the weighted linking matrix and handles teleportation implicitly.

As a result, all three PageRank experiments completed in a few seconds on the test machine.

The runtimes reported here are measurements from this particular MATLAB execution and should not be interpreted as hardware-independent performance guarantees.

The numerical experiment also illustrates an important tradeoff.

A larger damping factor places greater emphasis on the original hyperlink structure, but may require more iterations for convergence.

A smaller damping factor increases the contribution of teleportation and provides a stronger worst-case contraction bound.

Thus, the damping parameter affects both the mathematical model and its computational behavior.

Overall, our experiments confirm that PageRank can be computed efficiently on large sparse directed graphs using power iteration, while retaining the existence, uniqueness, and convergence guarantees developed earlier in this note.

7 Conclusion

In this note, we have explored the Perron–Frobenius theorem, beginning with the elementary properties of positive matrices and gradually developing its connections to stochastic matrices, Markov chains, and Google's PageRank algorithm.

Our mathematical investigation began with the question of whether a matrix whose entries are strictly positive must possess a positive eigenvalue with a corresponding strictly positive eigenvector.

By constructing an optimization problem over the standard simplex and using compactness, we established the existence of such an eigenvalue and eigenvector.

We subsequently proved that this eigenvalue is equal to the spectral radius, is algebraically simple, and strictly dominates every other eigenvalue in absolute value.

These properties distinguish positive matrices from arbitrary square matrices and provide the foundation for many important results in matrix theory.

We then introduced stochastic matrices and showed how their eigenvectors corresponding to the eigenvalue one describe stationary probability distributions.

This connection allowed us to interpret the evolution of a Markov chain through repeated matrix multiplication.

In particular, we saw that the Perron–Frobenius theorem provides sufficient conditions for the existence of a unique positive stationary distribution and convergence toward it.

The mathematical framework was then applied to Google's PageRank algorithm.

By representing hyperlinks as a directed graph and constructing a weighted transition matrix, we demonstrated that the problem of ranking web pages can be formulated as an eigenvalue problem.

We also explained how dangling pages, teleportation, and the damping factor modify the transition matrix to obtain desirable mathematical properties.

The resulting PageRank vector satisfies

$$Gr = r, \qquad r > 0, \qquad \mathbf{1}^T r = 1.$$

Our numerical implementation in MATLAB used the Stanford Web Graph dataset, containing 281,903 nodes and 2,312,497 directed edges.

The experiments demonstrated that PageRank can be computed efficiently using sparse matrix operations and power iteration.

We also observed that the damping factor has a substantial influence on the convergence rate, in agreement with our theoretical error estimate

$$\|x^{(k)} - r\|_1 \leq d^k \|x^{(0)} - r\|_1.$$

Beyond these mathematical and computational results, the history of the subject is equally fascinating.

The theory originated in the work of the German mathematicians Oskar Perron (1880–1975) and Ferdinand Georg Frobenius (1849–1917).

In 1907, Perron established fundamental spectral properties of matrices with strictly positive entries.

Frobenius subsequently developed the theory further, notably in 1912, extending important results to nonnegative matrices.

Their work helped establish a mathematical framework whose applications would eventually extend far beyond the problems that originally motivated its development.

Another remarkable historical connection concerns the Russian mathematician Andrey Markov (1856–1922).

Markov developed the theory of dependent random variables while investigating the extent to which classical probability results, including the law of large numbers, could remain valid without assuming independence.

In a particularly interesting application published in 1913, Markov investigated the arrangement of vowels and consonants in Alexander Pushkin's *Eugene Onegin*, a celebrated Russian novel in verse.

By examining the dependence between successive letters, he demonstrated how probabilistic reasoning could be applied to literary text.

It is a remarkable historical connection that mathematical ideas now used in search engines, communication systems, and data science were also developed and illustrated through the statistical analysis of poetry.

It is equally important to recognize the limitations of our PageRank model.

Although the original PageRank method was described in publicly available research, the complete ranking algorithms used by Google today are not publicly disclosed.

Google's modern search systems consider many additional signals and have evolved considerably since the original PageRank model.

Consequently, the algorithm implemented in this note should be understood as a mathematical reconstruction of the classical PageRank framework, based on its publicly documented principles.

It is not intended to reproduce the complete proprietary ranking system used by Google.

Nevertheless, this simplified model captures an important mathematical idea: the significance of a node in a network can be determined recursively from the significance of the nodes connected to it.

Finally, it is worth emphasizing that the theory of positive and nonnegative matrices is far from a closed chapter of mathematics.

Research in this field continues, including investigations involving spectral graph theory, nonnegative tensors, nonlinear eigenvalue problems, large-scale numerical computation, and applications in data science.

The theory is especially fascinating because seemingly elementary assumptions about matrix entries can lead to remarkably strong conclusions about eigenvalues, eigenvectors, stability, and long-term behavior.

What began as a mathematical investigation into the spectral properties of special classes of matrices has developed into a powerful collection of ideas with applications across mathematics and computation.

The Perron–Frobenius theorem therefore illustrates a broader lesson in mathematics: abstract theoretical results, sometimes developed long before their practical applications are imagined, can eventually become essential tools for understanding and solving problems in the modern world.

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